

电子科技大学
格拉斯哥学院
Glasgow College, UESTC

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Research on Retinal Vessel Segmentation Based on nnU-Net

Student:Chen Yichi

GUID:[redacted]

1st Supervisor: Keep blank for Moodle submission but put the name before hardcopy submission******

Abstract

This thesis investigates retinal vessel segmentation based on the nnU-Net framework. It focuses on three closely related problems: insufficient recovery of thin vessels, limited use of cross-level information, and unstable structural continuity. Under a unified experimental protocol, the study progressively examines both structural modification and result-refinement strategies. The early exploration stage evaluates input enhancement and loss adjustment, and the results suggest that extra input-side preprocessing does not provide clear gain on top of nnU-Net's existing preprocessing and training pipeline. The later main-experiment stage therefore shifts the focus back to the network structure itself. Starting from a unified nnU-Net baseline, the study proceeds through PA-Net module migration, LPT-based mainline screening, deformable sampling, multi-scale modeling, positional attention, and continuous geometric trajectory modeling, and eventually forms a final solution centered on MS-DSConv + CA with light post-processing. Under five-fold cross-validation on DRIVE, the final scheme achieves an F1-score of 0.8027, which is 0.0136 higher than the later-stage nnU-Net baseline result of 0.7891. These results indicate that structural improvement built around the continuous geometric trajectory of slender targets can enhance thin-vessel recovery and overall structural representation, while reaching a more suitable balance between vessel detection and false-positive control.

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1 Introduction

1.1 Research Background and Significance

The retina is one of the few tissues in the human body where microvascular status can be observed directly through non-invasive imaging. Because of this, the vascular morphology in fundus images has long been used in disease screening, computer-aided diagnosis, and clinical assessment [1, 2]. In practice, diabetic retinopathy, hypertension, arteriosclerosis, and some cardio-cerebrovascular diseases are often associated with changes in vessel calibre, branching pattern, tortuosity, and continuity. Reliable retinal vessel segmentation therefore supports later steps such as vessel width measurement, artery-vein analysis, lesion localization, and quantitative fundus evaluation, while also improving the automation and stability of ophthalmic diagnostic systems [3, 4].

Even so, retinal vessel segmentation is not an easy problem. Retinal vessels are slender, curved, and distributed across multiple scales. Peripheral micro-vessels are especially difficult because they are narrow, low in contrast, and easily affected by imaging noise, uneven illumination, and background texture interference [1, 2]. Bifurcations, crossings, and locally disconnected regions make the task harder still. When continuity and topological consistency are not preserved well, the reliability of subsequent quantitative analysis is reduced. For this reason, improving thin-vessel detection while maintaining structural continuity remains one of the central concerns in this task.

Deep learning has become the dominant paradigm in medical image segmentation, and encoder-decoder networks are now widely used in this area. U-Net and its variants, in particular, have shown strong performance because of their end-to-end learning framework and pixel-level prediction ability [5–7]. On top of this line of work, nnU-Net introduced a self-configuring pipeline that adapts preprocessing, network configuration, training strategy, and post-processing to specific biomedical segmentation tasks [8]. That design greatly reduces the amount of manual tuning required and makes nnU-Net a practical and credible baseline for retinal vessel segmentation, where both global semantic consistency and fine structural recovery matter.

Still, a general-purpose framework does not automatically solve the difficulties of a specific task. Retinal vessel segmentation depends not only on multi-scale feature extraction, but also on how well the model captures long-range dependency, key spatial locations, cross-level feature interaction, and geometric continuity. Previous studies suggest that attention mechanisms, feature fusion strategies, and more flexible sampling methods can help alleviate the limitations of conventional encoder-decoder models in recovering thin vessels, preserving boundaries, and maintaining structural continuity [9–12]. This is why it is worthwhile to explore task-oriented structural improvements on top of nnU-Net rather than treating the baseline as the final answer.

Against this background, this thesis takes nnU-Net as the basic framework and focuses on three closely related issues in retinal vessel segmentation: insufficient thin-vessel detection, limited cross-level information utilization, and inadequate structural continuity modeling. By progressively exploring different structural modules, sampling mechanisms, attention strategies, and output refinement methods, this study aims to provide experimental evidence and practical references for building a segmentation model that is better suited to retinal vessel analysis.

1.2 Related Work

Retinal vessel segmentation has been studied for many years and early work was largely based on image enhancement and hand-crafted rules. Common examples include matched filtering, thresholding, region growing, and morphological processing. These methods can work when vessel responses are clear and

imaging conditions are stable, but they are usually sensitive to image quality, noise, and parameter selection. Their weaknesses become more obvious in low-contrast regions, complex backgrounds, and thin peripheral vessels [1, 2]. As public datasets and evaluation protocols became more established, machine-learning and ensemble-classification methods were introduced to improve vessel-background discrimination, although this line of work still depended heavily on manually designed features [3, 4].

Deep learning changed the field quite quickly. Convolutional neural networks can learn hierarchical representations directly from raw images and have shown much stronger adaptability in medical image segmentation. U-Net and related encoder-decoder architectures became especially influential because they combine semantic feature extraction during downsampling with spatial detail recovery through skip connections [5–7]. In retinal vessel segmentation, these models soon replaced most traditional approaches and became the basis for later work [13–15].

That said, standard CNNs and the original U-Net do not fully solve the problem. Retinal vessels are slender, curved, and spread across multiple scales rather than appearing as regular compact targets. With local convolution alone, a model may segment major vessels reasonably well but still miss thin vessels, weak-response regions, and difficult bifurcations. Conventional encoder-decoder architectures also rely heavily on fixed skip connections, which limits their ability to model long-range dependency, key spatial locations, and cross-level structural continuity. This is why further task-specific refinement remains necessary even when the U-Net framework already performs well.

Many later studies kept the U-Net backbone and modified it more selectively. Some focused on the connection pattern between encoder and decoder or on multi-scale feature fusion. UNet++, for example, used nested skip connections to reduce the semantic gap between encoder and decoder features [16]. Others introduced attention mechanisms into the network. Attention U-Net is a representative case, using attention gates to suppress irrelevant responses and highlight target regions [10]. These studies do not really replace U-Net. Instead, they show where the original framework still needs help, especially in feature fusion, spatial selection, and structural recovery.

This matters even more in retinal vessel segmentation. Vessels are continuous, directional structures, and local convolution alone does not make good use of features that are far apart spatially but still related topologically. Weak vessel responses often appear in exactly these places. If the model handles them poorly, a vessel that should remain continuous is segmented into several disconnected pieces. For this reason, attention mechanisms, multi-scale fusion, and more flexible sampling strategies have gradually become common lines of work in this area.

nnU-Net pushed the field in a somewhat different direction by making the overall pipeline more standardized. It is better understood as a task-adaptive configuration framework than as a single new network, since it automatically adjusts preprocessing, network topology, training plans, and post-processing for a given dataset [8]. Because it has shown strong robustness across many medical image segmentation tasks and is not easy to surpass, it is now widely treated as a strong baseline. That is also why this thesis takes nnU-Net as its starting point: if an improvement remains effective on top of nnU-Net, it is more likely to be meaningful.

Recent work on retinal vessel segmentation has also moved more directly toward feature fusion and sampling design. PA-Net is a representative example. It introduces the Lightweight Parallel Transformer (LPT) and the Adaptive Vascular Feature Fusion Module (AFFM) into a U-Net-style backbone and combines them with a preprocessing pipeline designed for fundus images [9]. The aim is to improve both global structural modeling and multi-level vascular feature fusion. Work of this kind suggests that a standard encoder-decoder structure alone is often not enough for stable recovery of thin vessels, especially when key locations and

structural continuity need to be modeled more carefully.

Besides architectural changes, recent studies have paid more attention to the topological consistency of slender structures. Deep supervision can strengthen multi-scale feature learning through auxiliary outputs at multiple levels [17]. Deformable attention and modulation mechanisms increase spatial flexibility by learning sampling offsets and weights [11, 18]. Coordinate Attention supplements positional representation through direction-aware encoding [12], while Dynamic Snake Convolution tries to follow the geometry of slender targets through continuous trajectory-based sampling [19]. Topology-related loss functions and output post-processing strategies have also been used to compensate for limits in structural modeling. As a result, the question is no longer simply whether vessels can be segmented, but how thin vessels can be preserved more reliably, how continuity can be maintained, and how false positives can be kept under control.

Current research has already provided a substantial methodological basis for retinal vessel segmentation. Even so, there is still room for improvement in thin-vessel detection, cross-level information use, preservation of topological continuity, and the balance among evaluation metrics. For that reason, this thesis adopts nnU-Net as a unified baseline framework and further explores model designs and optimization strategies that are better suited to retinal vessel segmentation.

1.3 Major Challenges in Retinal Vessel Segmentation

Retinal vessel segmentation is often described as a pixel-level classification task, but the real difficulty lies in preserving fine vascular structure under non-ideal imaging conditions. Compared with the segmentation of organs or compact lesions, it places a heavier demand on delicate branch recovery and on the preservation of vascular continuity. In practice, the main difficulties tend to appear together, across image quality, feature representation, and metric trade-offs.

One source of difficulty comes from the vessels themselves. Retinal vessels are slender, curved, and distributed across multiple scales. Major vessels are usually easier to detect, but peripheral micro-vessels, weak responses, and complex bifurcations are much less stable. These regions are narrow, low in contrast, and easily mixed with background texture [1, 2]. When they are not recovered well, the result is often more than slight under-segmentation. Local discontinuities appear, fine branches disappear, and parts of a vessel tree that should remain continuous become fragmented.

Imaging variation makes the problem harder still. Fundus images acquired with different devices, under different settings, or from different patients can differ substantially in brightness, contrast, noise level, and background appearance [3, 4]. Public retinal datasets are also limited in number, which makes generalization harder to assess. Even within a single dataset, optic-disc highlights, lesion reflections, local shadows, and color variation may distort vessel responses. In practice, the model is asked to separate fine vessels from a visually cluttered background rather than from a clean and regular target distribution.

Another difficulty lies in feature representation. Convolutional neural networks are effective at local texture extraction, but long-range dependency, key spatial locations, and cross-level structural continuity are harder to model. Encoder-decoder architectures can recover part of the lost spatial detail through skip connections, yet they may still preserve high-level semantics more readily than fine low-level structure. The result is familiar in retinal vessel segmentation: local vessel evidence may be present, while the overall continuity of the vascular tree remains weak. This is one reason why cross-level interaction and structure-aware spatial modeling continue to receive attention [11, 12, 18].

Evaluation brings its own tension. A model that becomes more sensitive to thin vessels and weak responses often improves recall, but it may also mistake noise or background texture for vessels and lose precision. A more conservative model can produce cleaner outputs, yet it may also miss fine branches. For

this task, no single metric is enough. Sensitivity, precision, specificity, and F1-score usually need to be considered together.

There is also a practical problem. Even on top of a strong baseline such as nnU-Net, stable improvement is difficult. nnU-Net already provides a mature preprocessing, training, and inference pipeline [8]. Meaningful gains therefore usually come from changes that address these task-specific difficulties directly, rather than from superficial module stacking or minor parameter tuning. Recent work has largely moved in this direction, especially in structure-aware spatial modeling, the use of cross-level information, and output refinement [9, 11, 12, 19].

In the end, the recurring issues are fairly consistent: thin vessels and weak responses are hard to recover stably, robustness remains vulnerable under complex imaging conditions, local detail and global structure are difficult to preserve at the same time, and stronger detection still comes with the risk of more false positives. The later discussion of model design and optimization in this thesis is organized largely around these issues.

1.4 Main Research Content and Technical Route

This thesis takes nnU-Net as a unified baseline framework and studies how retinal vessel segmentation can be improved under the difficulties discussed above. The main concerns are thin-vessel recovery, the use of cross-level information, and the preservation of structural continuity. Rather than moving outside a strong baseline and building a completely new network from scratch, the study compares a series of modifications under a unified experimental setting and uses those comparisons to identify designs and optimization strategies that are better suited to the present task.

The work begins with a baseline and a set of preliminary experiments on input enhancement and loss-function adjustment. These early experiments are meant to clarify how much additional gain can still be obtained on top of nnU-Net's existing preprocessing and training pipeline. After that, the later experiments are conducted on a unified dataset under a consistent training protocol, so that differences among models can be interpreted with fewer confounding factors [8].

From there, the focus shifts to the network structure itself. Relevant design ideas are first transferred and adapted to the current framework, and their actual contribution is then examined through ablation. As the study moves forward, the emphasis turns more clearly toward high-level feature modeling and cross-level interaction, with particular attention to spatial sampling, multi-scale feature use, and representations that better match slender vascular structures [9, 11, 12, 18, 19]. The technical route is therefore shaped less by module accumulation than by repeated comparison, selection, and adjustment.

The thesis also includes supplementary exploration at the output and optimization levels, including post-processing and topology-related constraints. These parts are not introduced simply to push one metric higher. They are meant to seek a more workable balance among thin-vessel recovery, structural continuity, and false-positive control. The later chapters follow this progression and present the theoretical background, model implementation, and experimental analysis accordingly.

1.5 Organization of the Thesis

This thesis is organized into five chapters. Chapter 1 introduces the research background and significance of retinal vessel segmentation, reviews the related literature, outlines the main challenges of the task, and presents the main research content and technical route of this work. Chapter 2 provides the theoretical foundations and related methods, including the retinal vessel segmentation task itself, the basic principles of U-Net and nnU-Net, and the methods involved in the later model improvements. Chapter 3 focuses on the

improved model design and implementation. Following the main experimental line of the thesis, it explains the establishment of the baseline, the migration and screening of candidate structures, the main structural improvements, and the process of result refinement. Chapter 4 reports the experimental results and analysis, comparing the models at different stages and discussing model performance and the structural evolution path in light of the changes in key metrics. Chapter 5 concludes the thesis, summarizes the main work, and discusses the limitations of the current study together with possible directions for further improvement.

2 Theoretical Foundations and Related Methods

2.1 Overview of Retinal Vessel Segmentation

Retinal vessel segmentation aims to extract the vascular tree from fundus images and to produce vessel masks aligned with the original image. Formally, it is usually treated as a pixel-level classification problem, with each pixel assigned to either vessel or background. In practice, though, it is more than a foreground extraction task. The output is often used in later analyses such as vessel-width measurement, artery-vein analysis, lesion localization, and quantitative fundus assessment. For that reason, the value of a segmentation result depends not only on pixel-level accuracy, but also on whether continuity and branch completeness are preserved [1, 2].

Most current studies take a color fundus image as input and produce a vessel mask with the same spatial layout as the original image. The mask is usually obtained through manual annotation or expert revision and serves as the reference during training and evaluation. This task has an inherent class-imbalance problem because vessel pixels occupy only a small portion of the image, while the background dominates most of the field. Retinal vessels are also slender, curved, highly branched, and distributed across multiple scales. A useful model therefore needs to capture local texture cues without disrupting the global tree-like structure.

Commonly used public datasets include DRIVE, STARE, CHASE_DB1, and HRF [1, 2, 4, 20]. These datasets provide a familiar testing basis for retinal vessel segmentation research, but they are not identical in imaging device, resolution, field of view, pathological presentation, or annotation style. For that reason, results reported in the literature still need to be understood in light of the specific dataset conditions under which they were obtained.

Evaluation usually relies on a combination of sensitivity, specificity, accuracy, AUC, precision, and F1-score [1, 2]. Sensitivity is more informative for thin vessels and weak responses, AUC reflects overall discrimination capability, and F1-score is useful for judging the balance between vessel detection and false-positive control. Precision and specificity, by contrast, say more directly how well false positives are suppressed. Because this thesis is particularly concerned with vessel recovery, the later analysis places more weight on changes in sensitivity, AUC, and F1-score, while still using precision and specificity to check whether gains come at the cost of unstable false-positive behavior.

Taken together, the task imposes several demands at once. A model should recover thin vessels and complex bifurcations as completely as possible, reduce missed detections, avoid mistaking background texture or noise for vessels, and preserve the spatial continuity of the vascular tree. Much of the later work on network structure, feature interaction, and result refinement can be read as a direct response to these requirements.

2.2 Basic Principles of U-Net and nnU-Net

U-Net is one of the most influential encoder-decoder architectures in medical image segmentation [5]. By the time it was introduced, convolutional neural networks had already shown strong performance in image

classification, but medical image analysis was concerned much more with pixel-level localization than with single class labels. Large annotated datasets were also often unavailable in medical imaging, which made it unrealistic to transfer natural-image classification networks directly. What made U-Net important was that it brought contextual semantic modeling and precise spatial localization into the same end-to-end framework, so that segmentation tasks could still be handled reliably under limited annotation.

Structurally, U-Net consists of a contracting path and an expanding path. The contracting path gradually enlarges the receptive field through repeated convolution and downsampling so that higher-level semantic information can be extracted. The expanding path then restores spatial resolution through upsampling so that features return to the scale of the input image [5]. The key element in this design is the skip connection, which transfers shallow encoder features directly to the corresponding decoder stages and helps recover spatial details that would otherwise be lost during upsampling. This matters in medical image segmentation because boundaries, fine structures, and local textures are often preserved mainly in shallow features, whereas deeper features carry stronger semantic meaning at the cost of precise localization. A schematic illustration of the overall U-Net architecture is shown in Fig. 1.

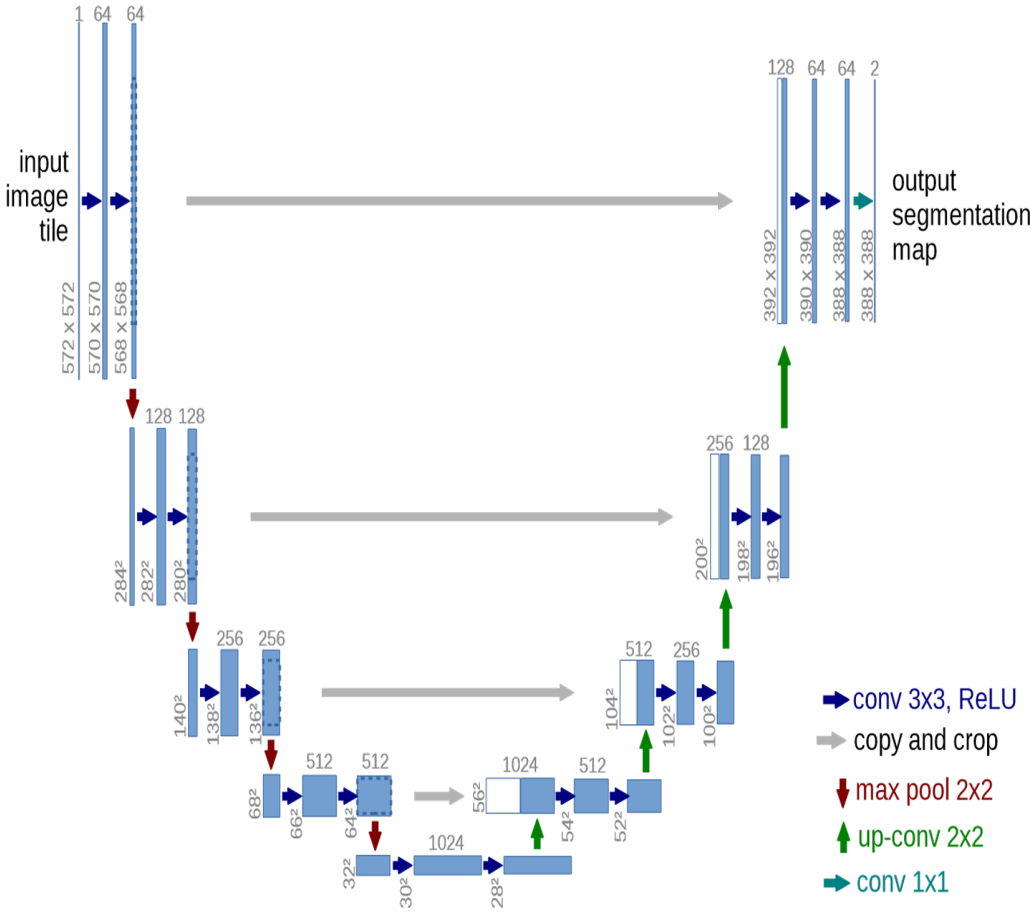


Figure 1: Schematic illustration of the classic U-Net architecture. The left side is the encoder path, which progressively extracts higher-level semantic features through convolution and downsampling; the right side is the decoder path, which gradually restores spatial resolution through upsampling; and the skip connections transmit shallow-detail information to the corresponding decoding stages.

Because of this complementarity between encoder, decoder, and skip connections, U-Net has shown strong adaptability in medical image segmentation. Downsampling provides relatively stable contextual information, which reduces the risk that decisions are made from local texture alone, while cross-level fusion helps preserve boundaries and fine details. For a task such as retinal vessel segmentation, where both fine

local recovery and global structural continuity matter, this design logic remains fundamental.

After U-Net, many structural variants were proposed in medical image segmentation, but they often required considerable manual adjustment for different datasets, hardware conditions, and training strategies. nnU-Net was introduced in response to this practical difficulty [8]. Rather than being a completely new network architecture, nnU-Net is better understood as an adaptive configuration framework for medical image segmentation. It does not try to replace the backbone with a radically different design. Instead, it determines preprocessing, network configuration, training strategy, and post-processing from the characteristics of the dataset, so that the full segmentation pipeline becomes more standardized and reusable.

An important part of nnU-Net is the way it turns task-related information into analyzable dataset properties and then organizes the later pipeline around them. The original work introduced ideas similar to a data fingerprint and a pipeline fingerprint: characteristics such as voxel or pixel spacing, image size, class distribution, and target-region size are first summarized, and these summaries are then used to determine more suitable network configurations and training plans [8]. In other words, nnU-Net does not begin with a fixed network and then force every task to fit it. It starts by characterizing the dataset and then builds the segmentation process around that characterization.

In practice, nnU-Net automatically organizes several key components of the pipeline, including voxel or pixel spacing handling, image normalization, patch-size selection, batch-size adjustment, network depth and convolution-kernel configuration, and the generation and comparison of different candidate models such as 2D U-Net, 3D full-resolution U-Net, and cascaded 3D U-Net [8]. During training, it also commonly employs deep supervision through auxiliary outputs at multiple levels in order to strengthen multi-scale feature learning [8, 17]. After training, it then uses validation results to choose among models or model ensembles and applies necessary post-processing to the predicted outputs, for example by deciding whether only the largest connected component should be retained. Its advantage therefore lies not only in the network body itself, but in the unified organization of the entire segmentation workflow, from preprocessing and training to validation, inference, and post-processing.

This turns many engineering choices that once depended heavily on experience into a more systematic set of rules, allowing researchers to focus more on the task itself rather than on repeated trial-and-error configuration. Because nnU-Net has shown strong stability across a wide range of medical image segmentation tasks, it is not only a common baseline but also a reliable starting point for later structural improvement studies. That matters in the present thesis, since the goal is not to search for a new starting point, but to examine whether structural modifications can produce real gains on top of a strong and stable unified framework.

For this thesis, U-Net provides the basic structural paradigm of medical image segmentation, whereas nnU-Net provides a unified implementation and configuration framework built on top of that paradigm. The former explains why encoder-decoder structure and skip connections became the basic solution in medical image segmentation, while the latter explains why later improvement studies can be carried out on a more stable and interpretable basis. For this reason, all later model design and experimental analysis in this thesis are developed on top of nnU-Net.

2.3 Attention Mechanisms and Feature Fusion Methods

In medical image segmentation, convolution alone is often not enough to handle local detail, long-range dependency, and cross-level semantic information at the same time. This limitation becomes more obvious in retinal vessel segmentation, where the target contains both dense local texture and an overall structure that is continuous, highly branched, and directionally variable. The model must therefore do two things at once: focus more accurately on information that is truly relevant to the target and organize feature representations

more effectively across different levels. Attention mechanisms and feature fusion methods have grown out of this need.

The basic idea of attention is to assign greater weight to more important locations, channels, or structures during feature representation, thereby suppressing irrelevant background responses. Compared with approaches that rely only on local convolution, attention can improve the model's ability to select task-relevant regions. A representative example is Attention U-Net, which introduces attention gates into the U-Net structure and filters the features passed from the encoder to the decoder so that the model can make better use of information related to the target structure [10]. The point is not simply to strengthen target response, but also to reduce interference from irrelevant regions during decoding.

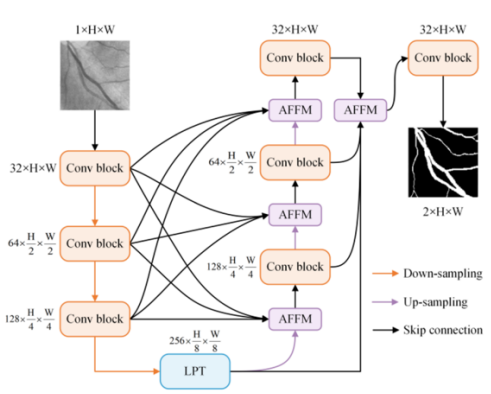
Beyond local attention, recent studies have increasingly focused on broader feature dependency, especially through self-attention and Transformer-like structures in segmentation. Their common feature is that they do not restrict feature interaction strictly to local convolutional neighborhoods, but instead attempt to establish associations across a larger spatial range. For retinal vessel segmentation, this is particularly attractive because the direction, bifurcation, and connectivity of slender vessels cannot be fully characterized by local texture alone. This is one reason why hybrid architectures that combine convolution and attention have become more common. They are usually intended to integrate global information more effectively in high-level semantic space without giving up too much local detail [9, 11, 21, 22].

Another closely related line of work is feature fusion. Although encoder-decoder architectures already provide a natural way to transfer information across levels, how to combine shallow detail with deep semantic information effectively remains unresolved. Standard U-Net relies mainly on skip connections for this purpose, while later work has paid more attention to whether a semantic gap exists between different feature levels and whether that gap affects recovery in the decoder. UNet++ is a representative example of this direction. It reconstructs the connection pattern between encoder and decoder through nested skip connections in order to reduce the semantic difference between different levels [16]. The broader lesson is that feature fusion is not simply a matter of concatenating multi-level features, but of improving the quality of the fused representation itself.

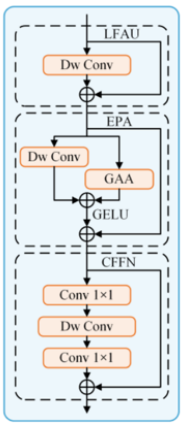
In retinal vessel segmentation, the importance of feature fusion is especially evident in the recovery of fine vessels and complex bifurcations. Deep features usually carry more stable semantic information, while shallow features preserve more boundary, texture, and local geometric detail. If the fusion strategy is too simple, the model may still achieve strong overall semantic judgment but fail to maintain sufficient resolution around fine structures. Many later methods therefore move beyond fixed cross-level connections and use weighting, selection, or adaptive reorganization so that different levels can contribute to decoding in a more targeted way.

Among the methods more directly related to the later work in this thesis, PA-Net introduced the Lightweight Parallel Transformer (LPT) and the Adaptive Vascular Feature Fusion Module (AFFM) [9]. LPT combines local feature extraction with broader dependency modeling in order to strengthen structural representation in high-level semantic space, whereas AFFM focuses on multi-level feature fusion in the decoder and aims to reduce the loss of vascular information caused by downsampling. The former is more closely related to attention and high-level structure modeling, while the latter relates more directly to feature fusion. The overall PA-Net architecture together with these related modules is illustrated in Fig. 2.

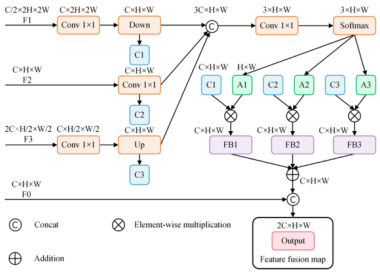
In the later chapters, attention mechanisms and feature fusion do not appear as two separate side topics. Together, they provide much of the methodological basis for the model improvements that follow.



(a) Overall PA-Net architecture



(b) LPT module



(c) AFFM module

Figure 2: PA-Net and two representative modules closely related to the later improvements in this thesis. The overall PA-Net structure shows where LPT and AFFM are located in the network; LPT is intended to strengthen structural modeling in high-level semantic space, whereas AFFM is designed to improve the effectiveness of multi-level feature fusion during decoding.

2.4 Deformable Sampling and Multi-scale Modeling

In standard convolution or fixed-receptive-field modeling, feature extraction is usually carried out on a regular grid, with sampling locations predetermined by the kernel shape. That works well for many targets, but it is often not flexible enough for structures that are slender, curved, and variable in direction. Retinal vessels fall squarely into this category. They are not regular block-like regions or rigid structures with clear boundaries, but elongated structures that extend, bifurcate, and change width across space. For such targets, a sampling process that stays at preset locations often struggles to follow the actual geometry closely enough.

The basic idea of deformable sampling is to move beyond predefined regular sampling positions by learning offsets that let feature aggregation adapt to the target structure. This idea appears in methods such as deformable convolution and deformable attention [11,18]. If the reference position is denoted by p, the k-th sampling offset by Δp_k, and the corresponding weight by w_k, the aggregation can be written in simplified form as

$$y(p) = \sum_{k=1}^{K} w_k \, x(p + \Delta p_k),$$

where $x(\cdot)$ denotes the input feature and K is the number of sampled locations involved in aggregation. Compared with standard convolution or fixed attention windows, deformable sampling places more emphasis on where to sample. By adjusting sampling positions dynamically, it allows the model to obtain more targeted responses around curved boundaries, slender branches, and complex bifurcations. In retinal vessel segmentation, that matters because the continuous trajectory of vessels is difficult to capture with only a small number of fixed grid locations.

In practice, deformable convolution mainly emphasizes learning more flexible sampling offsets within local neighborhoods, whereas deformable attention extends the same idea to a broader range of feature interaction. The former is typically used to improve how convolution adapts to irregular geometry, while the latter is more often used to determine the contribution of different sampling locations during feature aggregation. Despite these differences, both aim to reduce the mismatch introduced by fixed sampling and to make feature representation better aligned with the spatial distribution of the target itself. A schematic comparison between fixed sampling and deformable sampling is shown in Fig. 3.

If deformable sampling is mainly about flexibility in sampling position, multi-scale modeling is about

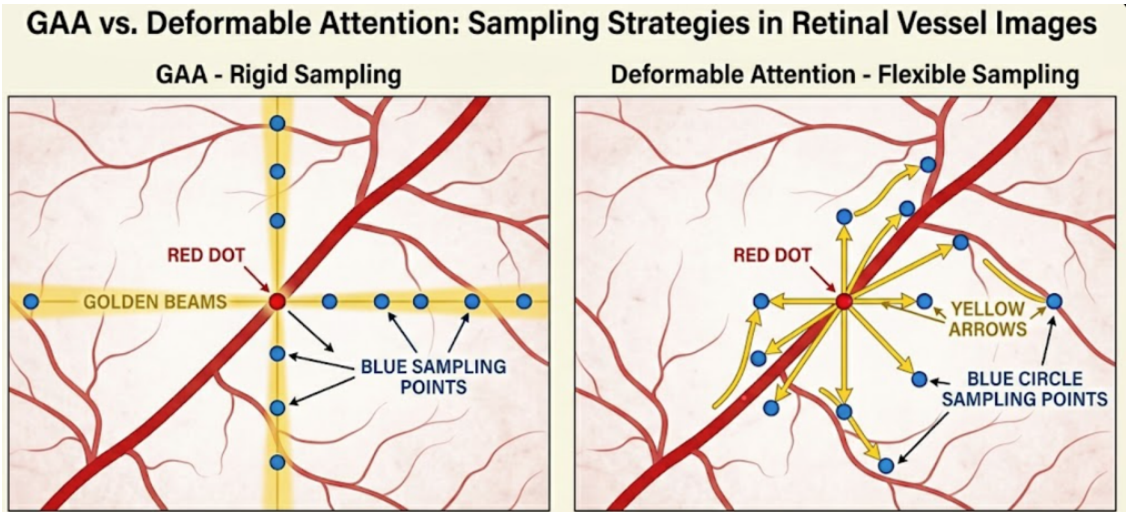


Figure 3: Schematic comparison between fixed sampling and deformable sampling. The left side shows a more regular fixed sampling pattern, whereas the right side shows deformable sampling with adaptive offsets around a reference position.

how information at different scales can participate more fully in representation. In retinal vessel segmentation, major vessel trunks, fine peripheral vessels, and complex bifurcations often correspond to different feature levels. Deep features are usually better suited to stable global judgment, while shallow features preserve more local boundary and texture detail. A single-scale representation therefore struggles to preserve both at once, which is why multi-scale modeling has become an important direction for improving vessel segmentation. If the input feature at scale l is denoted by x_l, the corresponding sampling offset by $\Delta p_{l,k}$, and the aggregation weight by $w_{l,k}$, then multi-scale deformable aggregation can be written as

$$y(p) = \sum_{l=1}^{L} \sum_{k=1}^{K} w_{l,k}\, x_l(p + \Delta p_{l,k}),$$

where L denotes the number of participating scales and K denotes the number of sampled locations at each scale. Compared with the earlier single-scale form, this expression emphasizes the joint participation of features from different levels around the same reference position.

Multi-scale modeling is not just a matter of feeding features of different resolutions into the network at once. More importantly, it is about how to build complementary relationships across scales. In encoder-decoder architectures, this often appears as cross-level transfer and fusion. In more advanced settings, it may take the form of cross-level sampling, cross-scale aggregation, or the explicit introduction of shallow-detail information into high-level representations. This is especially important for slender-structure segmentation, because such targets need global directional information to maintain continuity, while also requiring fine local detail to avoid losing peripheral vessels and complex bifurcations.

In methods more directly connected with the later work of this thesis, deformable sampling and multi-scale modeling are usually not treated as separate ideas. They are often combined instead: adaptive offsets increase the flexibility of sampling positions, while cross-level access and aggregation bring in information from different resolutions. Together, these two ideas make it easier for the model to follow the actual geometry of slender structures while also alleviating the loss of shallow detail in high-level semantic representations. The later improvements built around deformable attention, modulation, and multi-scale extension all rest on this methodological background.

2.5 Topology-oriented Methods for Slender Structures

For slender-structure segmentation tasks such as retinal vessel segmentation, pixel-level correctness does not necessarily imply structural correctness. Vascular networks are characterized by connectivity, bifurcation, and spatial extension. If a model produces strong local responses but fails to preserve continuity at the global level, the final result may still contain breaks, adhesions, or isolated noisy regions. This is why segmentation research has increasingly moved beyond local classification accuracy alone and begun to ask whether the predicted geometry and topology remain consistent with the real target structure.

From a structural modeling perspective, Dynamic Snake Convolution (DSConv) represents a line of work that places stronger emphasis on continuous geometric constraints [19]. Unlike regular convolution or discrete point-based sampling, this kind of method attempts to form a continuously unfolding trajectory between neighboring sampling positions, so that aggregation is no longer performed around a set of independent locations but instead follows the actual extension pattern of a slender target more closely. For structures such as vessels, which are curved, branched, and locally variable in width, the appeal of this design lies in its ability to preserve connectivity and directionality during feature extraction rather than merely reinforcing local responses.

More specifically, the key to DSConv is not simply that individual sampling points are movable, but that later sampling positions depend on the offsets of earlier ones. If the reference position is denoted by p, the horizontal step of the k-th sampled point by Δx_k, and the vertical increment by δy_k, then the sampling position along the snake-like trajectory can be written as

$$\tilde{p}_k = p + \left(\Delta x_k, \sum_{t=1}^{k} \delta y_t \right).$$

This expression highlights an important difference: unlike discrete point-based sampling, where each sampled position is treated independently, DSConv unfolds later positions from earlier ones and thereby forms a continuous trajectory. If local aggregation along that trajectory is further written as

$$y(p) = \sum_{k=1}^{K} w_k \, x(\tilde{p}_k),$$

then the core of DSConv becomes clearer: sampling points are first aligned more closely with the actual direction of the slender target, and feature aggregation is then performed along that continuous path. This is precisely why the method is particularly attractive for structures such as vessels, roads, and nerve fibers, where topology and continuity matter as much as local response.

Besides geometric constraints introduced during feature extraction, another common strategy is to apply post-processing to the predicted segmentation. In retinal vessel segmentation, model outputs often contain small and spatially isolated connected components that may come from background noise, lesion reflections, or local false positives. Post-processing does not alter the network body itself. Instead, it removes or corrects evidently unreasonable predictions according to the size, shape, or connectivity of the predicted components. The idea is relatively simple, but it acts directly on the topological cleanliness of the output and is therefore highly practical in real segmentation pipelines. For slender-structure tasks, its value is not limited to denoising. It also lies in reducing local predictions that are inconsistent with the overall connected form of the target.

Corresponding to structural constraints and output correction, another line of work introduces topology-

aware information through the loss function. clDice is a representative example because it incorporates connectivity constraints into the optimization target through skeleton-level overlap [23]. If the prediction is denoted by P and the ground truth by G, clDice is written as

$$\mathrm{clDice}(P,G) = \frac{2\,T_{\mathrm{prec}}(P,G)\,T_{\mathrm{sens}}(P,G)}{T_{\mathrm{prec}}(P,G) + T_{\mathrm{sens}}(P,G)},$$

where T_{prec} and T_{sens} denote topology-aware precision and topology-aware sensitivity defined on the basis of skeleton overlap. Unlike traditional losses that focus only on regional overlap, this type of method pays more attention to the correspondence between the predicted structure and the reference structure at the centerline or skeleton level. It is therefore particularly suitable for slender targets such as vessels, nerve fibers, and roads. In retinal vessel segmentation, the point is not only to encourage pixel-level agreement with the annotation, but also to preserve structural continuity at a higher level.

Taken together, topology-oriented methods for slender structures can be understood at three levels. Some introduce geometric constraints directly during feature extraction, some refine the output afterwards through connected-component rules, and others impose continuity-related constraints during training through additional losses. These methods act at different points in the pipeline: the first is closer to structural representation, the second is more concerned with output cleanup, and the third acts through the optimization objective. In the later work of this thesis, all three ideas appear in some form, although their roles and relative weight are not identical. What unites them is the same central concern: the segmentation result should not merely separate vessels from background, but should also remain structurally closer to the real vascular network.

For the later chapters of this thesis, this methodological background matters because it shows why structural continuity deserves to be treated as a problem in its own right. The later improvements built around continuous geometric modeling, post-processing, and topology-related losses are developed within this line of thinking.

2.6 Chapter Summary

This chapter has followed the experimental progression from early exploration to later structural optimization within the unified nnU-Net framework. The early experiments make one point clear: baseline training, input enhancement, and loss-function adjustment can influence the training process and some local metrics, but they do not truly resolve the insufficient recovery of thin vessels or the weak modeling of structural continuity. It is from this point that the study begins to move away from training-side adjustment and toward structural improvement of the network itself.

The later stages of the chapter begin with PA-Net module migration as a first round of structural exploration. Through full migration, single-module ablation, and the control experiment on AFFM and deep supervision, the thesis carries out an initial screening of how different modules adapt to the present framework. The overall result is that the complete PA-Net migration does not yield a better solution, AFFM does not provide stable gain, and LPT shows a clearer value for continued optimization. From there, the structural mainline gradually converges on LPT and continues through deformable sampling, modulation, multi-scale aggregation, positional attention, and continuous geometric trajectory modeling.

As structural continuity modeling becomes stronger, the model also begins to expose a corresponding problem in the form of false positives, even when SE, AUC, and F1-score improve. This is why the second half of the chapter turns to supplementary exploration from the perspectives of output refinement and loss-based constraints. The results show that area-threshold post-processing can correct local false positives more directly and provide a more stable overall gain, whereas clDice, although reasonable in direction, does

not produce a better overall result under the current experimental conditions. What is finally retained in this chapter is therefore not one isolated module, but a mainline solution formed through gradual screening together with its necessary output-side refinement strategy.

In that sense, this chapter is really about how the model was modified step by step. If Chapter 2 explains what these methods are, Chapter 3 explains how they are selected, combined, and implemented in the present work. The next chapter will examine the models formed here from the perspectives of overall results, metric trade-offs, and structural evolution.

3 Improved Model Design and Implementation

3.1 Overall Research Idea and Experimental Framework

Before turning to specific structural improvements, this thesis first establishes a relatively clear experimental framework so that the later models can be compared on a common basis. The overall strategy is not to test different modules under scattered settings, but to move from preliminary exploration toward a unified protocol and then carry out later structural improvement and result refinement under the same experimental conditions. In broad terms, the experiments fall into two stages: an early exploration stage, used to establish an initial reference and to examine routine training-side adjustments, and a later main-experiment stage, where subsequent structural modifications are compared under unified data conditions and training protocols.

The early exploration stage corresponds to the earliest control experiments. Its role is to establish an initial baseline and to observe how input enhancement and loss-function adjustment affect segmentation performance. At this stage, training and validation are based on multi-source samples collected from public fundus datasets, mainly STARE, CHASE_DB1, and HRF [2,4,20], following the existing split, while DRIVE is used as the test set so that the initial model behavior can be observed on a relatively unified target domain. To preserve as much original color information as possible, the input is organized in a three-channel separated form: the fundus image is first converted into the NIfTI format used by nnU-Net, and the red, green, and blue channels are then treated as separate inputs. The purpose here is to provide a somewhat broader preliminary reference before the later formal comparison, rather than to define the final solution directly.

The later main-experiment stage begins with the establishment of a unified protocol, and the focus correspondingly shifts from early feasibility assessment to structural improvement. Here, the DRIVE dataset is adopted as the experimental basis, together with a consistent data organization scheme and a five-fold cross-validation protocol. Unlike the earlier stage, these experiments no longer use three-channel color input, but instead rely on enhanced single-channel images. More specifically, the original fundus images are processed through grayscale transformation, normalization, adaptive histogram equalization, and gamma correction before being saved as single-channel NIfTI inputs. This preprocessing pipeline follows the general idea used in PA-Net [9] and is intended to improve the distinguishability of thin vessels and low-contrast regions while reducing irrelevant color information. Once the later baseline is established, all subsequent structural-improvement experiments are carried out under this unified benchmark so that data-condition differences are controlled as much as possible and horizontal comparison among structures remains grounded on the same basis.

All experiments are implemented on the nnU-Net v2 framework [8]. The early exploration stage is used mainly for rapid reference building and trend observation, and therefore retains the fixed split together with a relatively simplified comparison setting. The later main experiments uniformly adopt the 2D configuration of nnU-Net v2, with an input patch size of 512×512, a batch size of 2, 100 training epochs, SGD optimization, an initial learning rate of 0.001, and a Poly learning-rate decay schedule. Unless otherwise noted, all models

share the same data split, training plan, and inference procedure. Put differently, the main variables in the later experiments come from the network structure, loss function, or post-processing strategy itself rather than from changes in the protocol.

The evaluation strategy also differs slightly between the two stages. The early exploration stage mainly uses IoU, mIoU, Recall, Precision, Accuracy, mPA, and F1-score to establish an initial basis for comparison, whereas the later main-experiment stage uniformly adopts Sensitivity (SE), Specificity (SP), Accuracy (ACC), AUC, Precision (PR), and F1-score. Because the present task is more concerned with thin-vessel recovery and overall structural balance, the later analysis places greater weight on changes in SE, AUC, and F1-score, while using PR and SP to judge false-positive control. For the five-fold cross-validation results, conclusions are not drawn from any single fold in isolation. Predictions from all folds are aggregated before the final metrics are computed, reducing the fluctuation caused by any one split and improving consistency in model comparison.

3.2 Baseline Model and Preliminary Exploration

After establishing the overall experimental framework, this thesis begins with the early baseline experiments together with input enhancement and loss-function exploration around that baseline. This stage is not meant to define the final solution directly. It addresses a more basic question first: how much improvement can routine training-side adjustments actually bring to the present task. Since it still functions mainly as an early feasibility check, the existing training and validation split is retained, and DRIVE is used uniformly as the test set so that different settings can be compared under the same data conditions.

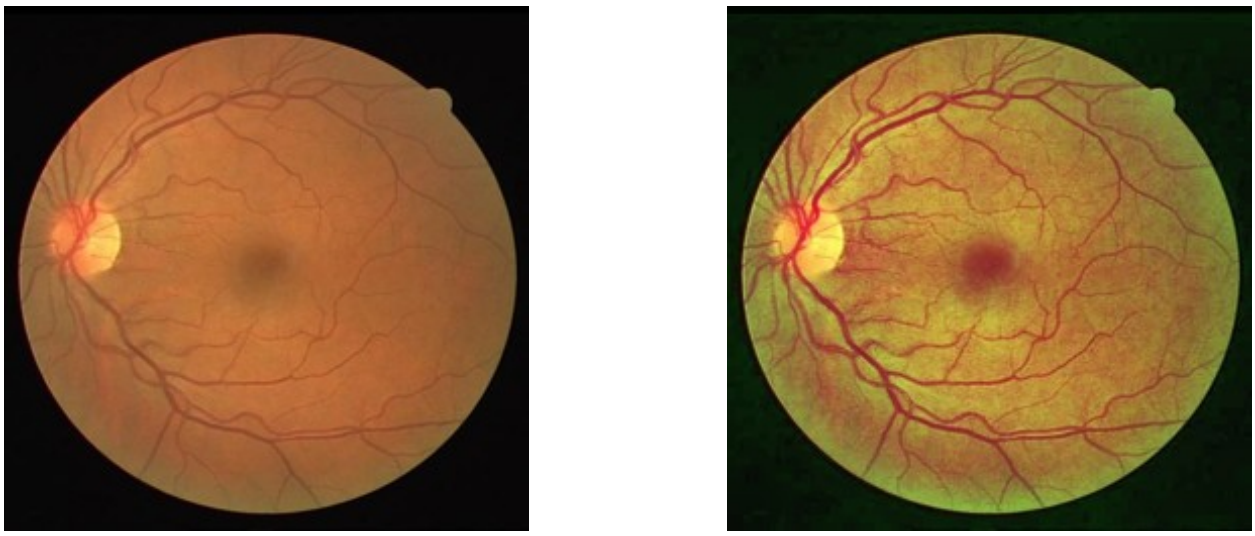
The first experiment adopts the standard nnU-Net baseline model [8] as the initial performance reference. It is trained with the original input form and standard training settings in order to see what level of segmentation performance the current framework can achieve before any additional structural modification is introduced. Standard nnU-Net already provides a relatively stable overall performance at this stage and can therefore serve as a reasonable starting point for later exploration.

The second experiment introduces CLAHE-based enhancement within the same framework to examine the effect of input enhancement on vessel segmentation. Thin vessels and low-contrast regions in fundus images are often not sufficiently clear, so moderate local contrast enhancement may improve the model's perception of these areas. Fig. 4 provides a direct comparison between the original fundus image and the CLAHE-preprocessed result. After CLAHE, local contrast becomes stronger and some fine-vessel regions appear more clearly, although part of the non-vessel texture and background response is also enhanced. CLAHE does help improve early-stage convergence, but its effect on final segmentation performance remains limited. For this early-stage setting, a more suitable explanation is not that such enhancement is meaningless, but that nnU-Net already provides a fairly mature internal preprocessing pipeline, so additional manual enhancement before the image enters the network does not bring the clear gain one might expect.

The third experiment further introduces Focal Loss on top of CLAHE so as to examine the effect of loss-function adjustment on hard samples and class imbalance. Compared with the baseline and with input enhancement alone, this setting leads to faster convergence, but the final performance does not improve further. Put differently, when the underlying structure remains unchanged, the influence of loss-function adjustment still falls mainly on the training process itself rather than on a stable improvement in the final result.

Table 1 summarizes the settings and representative results of these three early experiments.

Looking back at this stage, input enhancement and loss-function adjustment do affect the training process



(a) Original fundus image (b) CLAHE-preprocessed image

Figure 4: Comparison between the original fundus image and the CLAHE-preprocessed result. CLAHE enhances local contrast and makes some thin-vessel regions more visible, but it also strengthens part of the non-vessel texture and background response.

Table 1: Summary of preliminary exploration experiments

Experiment	Method	Training setup	Convergence	Representative result	Main observation
Exp1	nnU-Net baseline	Dice, 150 epochs	converged at epoch 69	F1 = 0.7924	established the initial baseline
Exp2	CLAHE + nnU-Net	Dice, 150 epochs	converged at epoch 60	F1 = 0.7827	improved early convergence, but limited final gain
Exp3	CLAHE + nnU-Net + Focal Loss	Focal Loss, 150 epochs	converged at epoch 53	F1 = 0.7802	faster convergence, but no further performance improvement

and produce changes in some local metrics, but they do not translate into sufficiently stable gains in the final result. In particular, the extra input-side preprocessing does not bring the clear benefit that might have been expected, which is consistent with the fact that nnU-Net already has a relatively complete internal preprocessing pipeline. For that reason, the focus of the later study shifts gradually from local training-side adjustment to structural improvement of the network itself.

3.3 PA-Net Module Migration and Ablation Experiments

After the preliminary exploration showed that extra input-side preprocessing did not bring clear gain and that loss-function adjustment also failed to produce stable improvement, the focus of this study shifted to the network structure itself. The first step at this stage was to draw on the design of PA-Net and attempt to migrate two of its core modules that are particularly relevant to retinal vessel segmentation into the nnU-Net framework, namely the Lightweight Parallel Transformer (LPT) and the Adaptive Vascular Feature Fusion Module (AFFM) [9]. These two modules were chosen because they correspond to the two issues that matter most for the later study: structural representation in high-level feature space and effective fusion of different feature levels during decoding.

From a functional perspective, LPT is placed at the bottleneck and is intended to strengthen the modeling of long-range dependency and global structural information, whereas AFFM replaces the standard skip-connection fusion pattern and is expected to reduce the loss of detail caused by downsampling through adaptive weighting. Since the basic principles of these modules have already been introduced in Chapter 2, the focus here is not on their internal mechanism as such, but on whether they can produce stable gains once migrated into nnU-Net under a unified protocol. At this point, the goal is not to identify the final model immediately, but to test whether the structural ideas behind PA-Net can actually hold in the present framework.

With this in mind, the thesis first constructs a full PA-Net migration model and then splits it into two single-module variants under the same experimental setting, namely LPT Only and AFFM Only, in order to determine which part of the complete migration is most responsible for the observed performance changes. The corresponding results are listed in Table 2.

Table 2: Results of PA-Net module migration and ablation experiments

Model	SE	SP	ACC	AUC	PR	F1-score
nnU-Net v2 Baseline	0.7627 ± 0.0446	0.9852 ± 0.0050	0.9659 ± 0.0045	0.8739 ± 0.0210	0.8238 ± 0.0646	0.7891 ± 0.0320
PA-Net (Full)	0.7211 ± 0.0763	0.9888 ± 0.0050	0.9659 ± 0.0059	0.8549 ± 0.0365	0.8570 ± 0.0621	0.7782 ± 0.0464
LPT Only	0.7597 ± 0.0492	0.9864 ± 0.0041	0.9669 ± 0.0044	0.8731 ± 0.0234	0.8361 ± 0.0546	0.7935 ± 0.0297
AFFM Only	0.7573 ± 0.0475	0.9868 ± 0.0039	0.9672 ± 0.0045	0.8720 ± 0.0227	0.8400 ± 0.0517	0.7942 ± 0.0300

Table 2 shows that the full PA-Net migration does not outperform the baseline. Although PA-Net (Full) improves PR, indicating a more conservative prediction pattern, its SE drops noticeably, and its final F1-score also falls below the baseline. The fully migrated model therefore tends to retain high-confidence vessel regions at the expense of detecting some thin vessels and weak-response areas. By contrast, the two single-module ablations remain more stable overall. LPT Only slightly improves PR and F1-score while largely maintaining SE, which suggests that stronger high-level structural modeling at the bottleneck may be useful. AFFM Only achieves a relatively high ACC and F1-score within this local comparison, but the result is still not strong enough to justify AFFM as the clear mainline direction for the later study.

At that point, the question narrows to AFFM itself. As a multi-level feature fusion module operating in the decoder, could its gain be affected by the default deep-supervision mechanism in nnU-Net? Since deep supervision also acts on multiple decoder outputs and reinforces multi-scale feature learning through auxiliary segmentation heads [8, 17], the thesis further constructs an AFFM w/o DS model by turning off deep supervision on top of AFFM Only, in order to examine whether a potential conflict exists between the two. If the main output loss is denoted by $\mathcal{L}_{\text{main}}$ and the s-th auxiliary loss by $\mathcal{L}_{\text{aux}}^{(s)}$, the total loss under deep supervision can be written as

$$\mathcal{L} = \mathcal{L}_{\text{main}} + \sum_{s=1}^{S} \lambda_s \mathcal{L}_{\text{aux}}^{(s)},$$

where λ_s denotes the weight of each auxiliary branch. The training mechanism is illustrated in Fig. 5.

The corresponding results are shown in Table 3.

Table 3: Comparison of AFFM with and without deep supervision

Model	SE	SP	ACC	AUC	PR	F1-score
AFFM Only (with DS)	0.7573 ± 0.0475	0.9868 ± 0.0039	0.9672 ± 0.0045	0.8720 ± 0.0227	0.8400 ± 0.0517	0.7942 ± 0.0300
AFFM w/o DS	0.7401 ± 0.0493	0.9880 ± 0.0038	0.9667 ± 0.0048	0.8640 ± 0.0236	0.8490 ± 0.0534	0.7883 ± 0.0308

As shown in Table 3, removing deep supervision leads to slight improvements in SP and PR but reduces SE, AUC, and F1-score, indicating a more conservative model overall. Turning off deep supervision reduces some false positives, but it also weakens the detection of thin vessels and weak responses. This means that simply disabling deep supervision does not improve the overall behavior of AFFM, and the earlier performance pattern cannot be explained away as a straightforward case of "deep-supervision conflict."

Taken together, the results of this section suggest that the individual design ideas in PA-Net do have some value, but directly migrating the complete structure into nnU-Net does not yield a better solution. AFFM shows some local promise, but its gain is not stable, and the control experiment with deep supervision does not reveal a clearer advantage. By contrast, LPT shows a more convincing value for further exploration at this stage. The importance of this section is therefore not that it identifies the final model, but that it

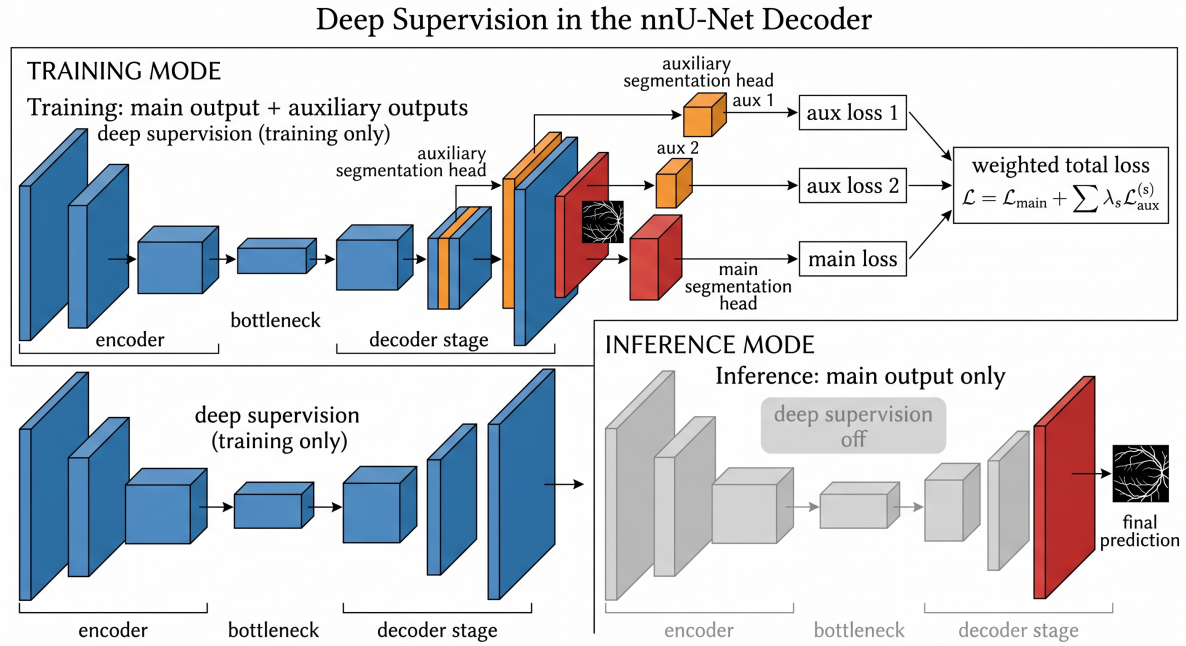


Figure 5: Schematic illustration of deep supervision in the nnU-Net decoder. During training, auxiliary segmentation heads are attached to multiple decoding stages and jointly contribute to the weighted loss, while only the main output is retained at inference time.

completes the first round of structural screening: directions worth pursuing are retained, while the next stage of improvement can move forward on a clearer basis around LPT.

3.4 Structural Improvements Based on LPT

After the first round of structural screening, the later improvement path of this thesis narrowed to the LPT module. Compared with the full PA-Net migration and the AFFM-centered exploration, LPT showed more stable value for continued optimization within the present framework. The point, then, is not that PA-Net as a whole was useless, but that the part most worth retaining for the later study was the one more directly related to high-level structural modeling. For that reason, this section moves away from decoder-side fusion and returns to the bottleneck LPT, tracing the later improvements built around its sampling mechanism and feature aggregation.

The first step in this round of improvement is deformable sampling. Drawing on the idea of deformable attention [11], the thesis replaces the original GAA form of LPT in PA-Net with a more flexible deformable sampling mechanism. Fixed sampling can provide relatively stable local feature aggregation, but it still adapts poorly to curved vessels, complex bifurcations, and fine peripheral branches. Deformable sampling, by contrast, allows the sampling positions to shift adaptively around a reference point, making it more likely that the true trajectory of slender structures can be preserved in high-level semantic space. Since Chapter 2 has already introduced the difference between fixed and deformable sampling, the emphasis here is not to restate the method, but to examine what this replacement actually changes in experiment.

To evaluate this replacement, three comparison models are constructed in succession: first, Deformable LPT (4 point) without AFFM, to judge whether the deformable replacement itself is effective; second, Deformable LPT with AFFM reintroduced under the same structure, to determine whether AFFM still needs to be retained in the new LPT form; and third, a version in which the number of sampled points is expanded from 4 point to 8 point, to observe the effect of sampling scale. The corresponding results are shown in Table 4.

Table 4: Results of deformable LPT-related experiments

Model	SE	SP	ACC	AUC	PR	F1-score
LPT Only	0.7597 ± 0.0492	0.9864 ± 0.0041	0.9669 ± 0.0044	0.8731 ± 0.0234	0.8361 ± 0.0546	0.7935 ± 0.0297
Deformable LPT (No AFFM, 4 point)	0.7634 ± 0.0540	0.9863 ± 0.0043	0.9672 ± 0.0044	0.8749 ± 0.0254	0.8371 ± 0.0511	0.7958 ± 0.0275
Deformable LPT + AFFM	0.7571 ± 0.0495	0.9867 ± 0.0042	0.9670 ± 0.0044	0.8719 ± 0.0235	0.8388 ± 0.0545	0.7933 ± 0.0301
Deformable LPT (No AFFM, 8 point)	0.7573 ± 0.0506	0.9869 ± 0.0038	0.9672 ± 0.0047	0.8721 ± 0.0241	0.8412 ± 0.0504	0.7946 ± 0.0296

As shown in Table 4, replacing LPT with deformable sampling without AFFM improves SE, AUC, and F1-score relative to LPT Only, and the 4 point setting gives the highest F1-score within this group. This suggests that a more flexible sampling mechanism at the bottleneck improves high-level vessel-structure modeling. Reintroducing AFFM does not lead to further improvement: F1-score drops from 0.7958 to 0.7933, which again suggests that AFFM does not provide stable gain in the present framework. The 8 point setting raises PR slightly, but SE, AUC, and F1-score all remain below the 4 point version. The later experiments therefore retain Deformable LPT (4 point) without AFFM as the new improvement baseline.

With that baseline in place, the thesis continues along the same line. The next step introduces a modulation mask within the single-scale deformable sampling framework, yielding Modulated Deformable (4 point). On top of that structure, the sampling range is further extended from the current layer to multiple encoder levels, so that cross-level access and aggregation can provide more high-resolution information, yielding MS-Modulated (4 point). In this chapter, MS-Modulated is therefore treated as the form obtained by adding multi-scale sampling to Modulated Deformable. Figures 6 and 7 illustrate these two steps.

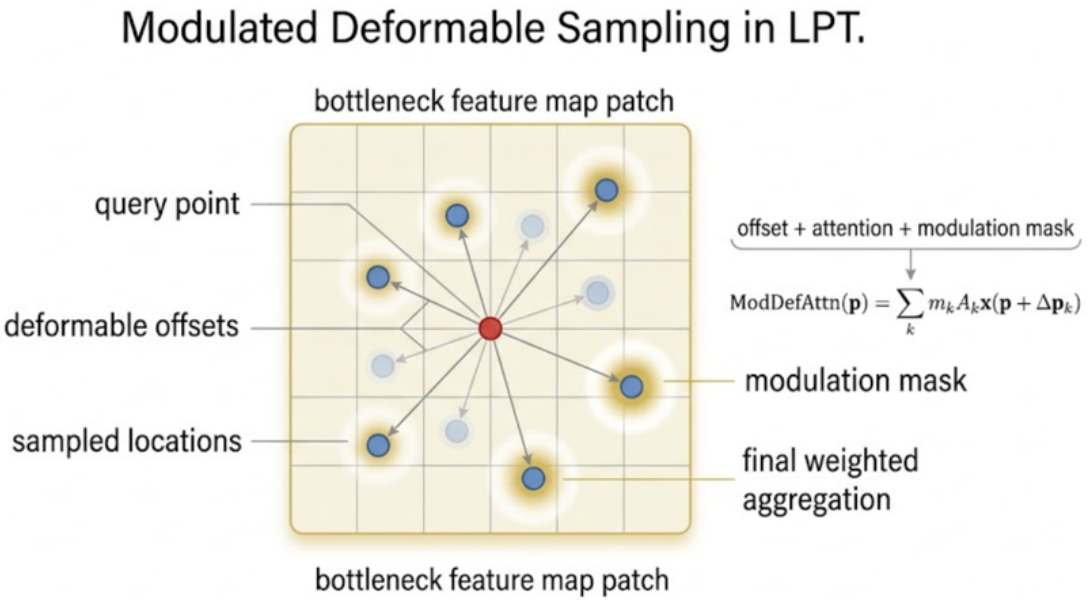


Figure 6: Schematic illustration of single-scale Modulated Deformable sampling. The query position accesses nearby locations through deformable offsets, and a modulation mask further adjusts the contribution of different sampled points.

On the basis of this progression, the thesis further constructs the Modulated Deformable (4 point) and MS-Modulated (4 point) models and compares them under the same experimental setting. The corresponding results are listed in Table 5.

Table 5 shows that Modulated Deformable (4 point) achieves the highest SE and AUC in this group, indicating that the modulation mask improves the detection of thin vessels and weak-response regions. Within single-scale deformable sampling, the additional selection mechanism does seem to alter the sampling distribution and push the model toward positions more closely related to vascular continuity. Once multi-scale

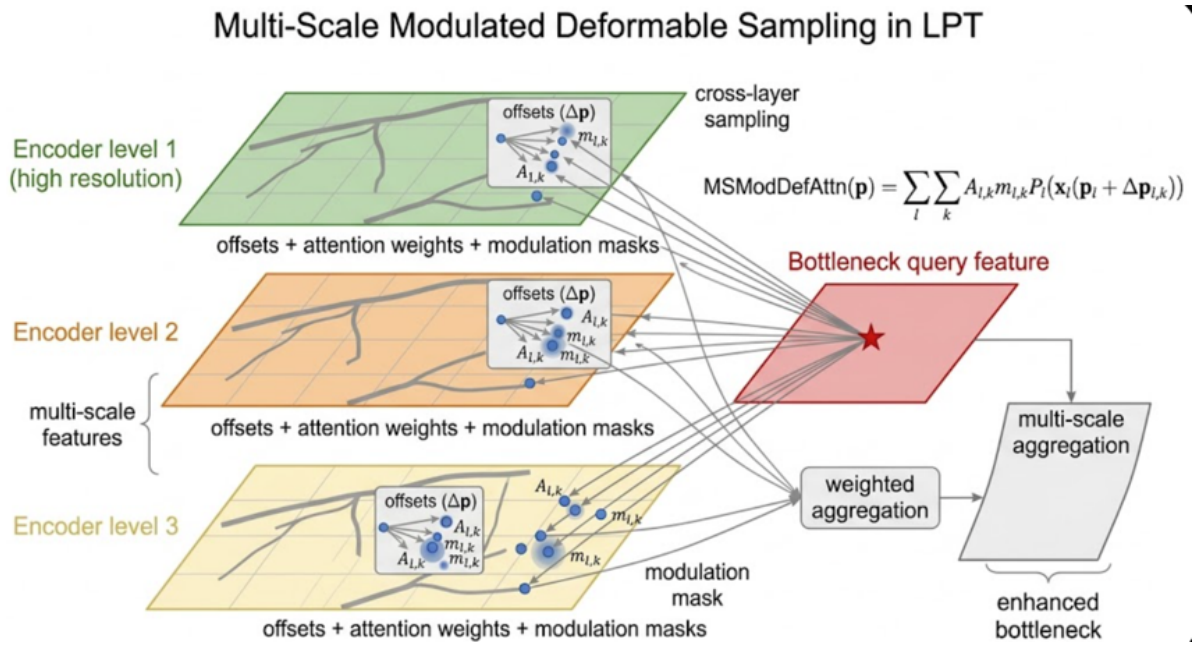


Figure 7: Schematic illustration of multi-scale Modulated Deformable sampling. Bottleneck features act as queries, access multi-scale encoder features across levels, and form an enhanced bottleneck representation through weighted aggregation.

Table 5: Results of Modulated and multi-scale deformable LPT experiments

Model	SE	SP	ACC	AUC	PR	F1-score
Deformable LPT (No AFFM, 4 point)	0.7634 ± 0.0540	0.9863 ± 0.0043	0.9672 ± 0.0044	0.8749 ± 0.0254	0.8371 ± 0.0511	0.7958 ± 0.0275
Deformable LPT (No AFFM, 8 point)	0.7573 ± 0.0506	0.9869 ± 0.0038	0.9672 ± 0.0047	0.8721 ± 0.0241	0.8412 ± 0.0504	0.7946 ± 0.0296
Modulated Deformable (4 point)	0.7660 ± 0.0511	0.9858 ± 0.0048	0.9669 ± 0.0046	0.8759 ± 0.0241	0.8324 ± 0.0560	0.7950 ± 0.0305
MS-Modulated (4 point)	0.7582 ± 0.0509	0.9870 ± 0.0038	0.9673 ± 0.0046	0.8726 ± 0.0242	0.8419 ± 0.0497	0.7954 ± 0.0283

sampling is introduced on top of this structure, MS-Modulated (4 point) no longer raises SE, but it performs better in SP, ACC, and PR. This suggests that cross-level multi-scale sampling helps suppress background noise and improve the overall prediction balance. Put differently, the move from Modulated Deformable to MS-Modulated shifts the emphasis from stronger detection toward richer cross-level feature integration.

Taken together, this section presents a clearer progression in the LPT line: deformable sampling first addresses the limitation of fixed sampling for slender vessels, modulation then strengthens key-position selection within the single-scale setting, and multi-scale sampling finally introduces richer feature support from different levels. Although the performance differences at this stage are still limited, the research line has already moved from asking whether a module can be migrated at all to asking how a more stable logic of structural improvement can be formed within a unified framework. On that basis, the next section continues along the multi-scale direction and further introduces attention enhancement and continuous geometric modeling.

3.5 Multi-scale Modeling and Attention Enhancement

At the end of the previous section, the improvement mainline had already reached the MS-Modulated stage. By that point, the model was able to access and aggregate multi-scale features across levels, but a new issue also became apparent. Although multi-scale sampling helps recover shallow detail and improves the overall prediction balance, cross-level aggregation alone is still not enough to fully strengthen the selective representation of vessel direction and spatial location. On that basis, the thesis first introduces Coordinate Attention (CA) into the MS-Modulated structure [12], with the aim of enhancing directional and positional modeling on top of the existing multi-scale design.

Unlike ordinary channel attention, Coordinate Attention preserves both channel relationships and spatial directional information. Its basic procedure is to aggregate features separately along the horizontal and vertical directions and then produce two direction-sensitive attention maps through a shared transformation. If the input feature is denoted by $X \in \mathbb{R}^{C\times H\times W}$, the output can be summarized as

$$Y_{\mathrm{CA}} = X \odot a_h \odot a_w,$$

where a_h and a_w denote the attention responses generated along the two spatial directions. For retinal vessel segmentation, this design is useful because it does not simply enhance all channels uniformly. Instead, it further emphasizes regions related to vessel direction, location, and continuity on top of the existing multi-scale aggregation result. In implementation, CA is placed at the entrance of the multi-scale LPT module: the bottleneck feature is first recalibrated in a direction-aware manner and is then sent to the later local feature extraction and multi-scale deformable aggregation stages. Figure 8 illustrates the combined structure of MS-Modulated and Coordinate Attention.

On this basis, the thesis uses MS-Modulated (4 point) as the direct reference model and further constructs MS-Modulated + CA in order to examine whether introducing explicit positional attention on top of cross-level multi-scale sampling can produce a more stable overall improvement. The corresponding results are shown in Table 6.

Table 6: Comparison between MS-Modulated and MS-Modulated + CA

Model	SE	SP	ACC	AUC	PR	F1-score
MS-Modulated (4 point)	0.7582	0.9870	0.9673	0.8726	0.8419	0.7954
MS-Modulated + CA	0.7697	0.9857	0.9673	0.8777	0.8324	0.7973

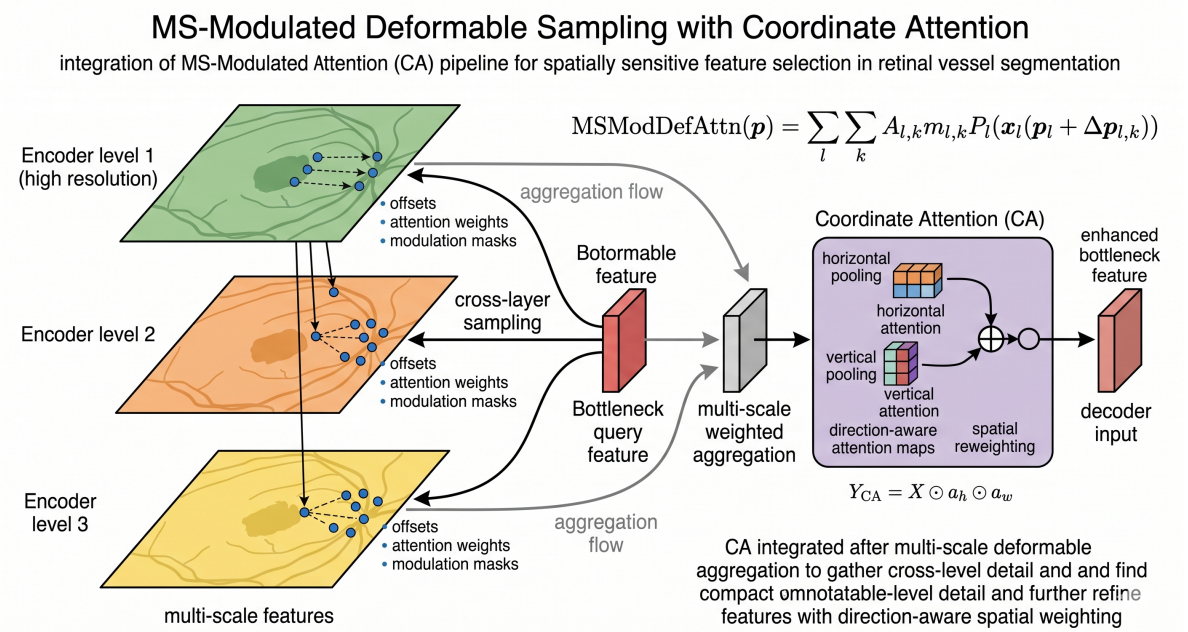


Figure 8: Structure of MS-Modulated combined with Coordinate Attention. Multi-scale deformable sampling first completes cross-level information aggregation, after which Coordinate Attention further recalibrates the feature response along spatial directions.

Table 6 shows that, after CA is introduced on top of MS-Modulated, the model improves further in SE, AUC, and F1-score, with SE reaching 0.7697, AUC reaching 0.8777, and F1-score increasing to 0.7973. This indicates that, once cross-level deformable sampling is already in place, explicit positional attention can further enhance the detection of thin vessels and weak responses while improving the overall segmentation quality. Compared with the previous section, MS-Modulated tends to improve SP, ACC, and PR, whereas the addition of CA leads to better SE, AUC, and F1-score. In that sense, CA is not simply pushing false positives down further. More importantly, it strengthens representations related to vessel direction and position on top of the existing multi-scale interaction.

Another issue becomes hard to ignore at this stage. Although multi-scale deformable sampling and Coordinate Attention have already strengthened cross-level interaction and spatial selection, the current sampling mechanism is still closer to a discrete point-based model: it can locate useful positions, but it does not yet describe the slender, continuous, and curved geometry of retinal vessels sufficiently well. This is why the later experiment introduces Dynamic Snake Convolution (DSConv) [19], with the intention of imposing continuous trajectory constraints among sampled points so that the sampling process can better follow the actual course of retinal vessels. This step does not replace ordinary convolution throughout the whole network. Instead, DSConv is introduced into the multi-scale sampling branch of the bottleneck LPT module, replacing the discrete cross-level deformable sampling used in the previous MS-Modulated + CA stage.

Unlike the earlier discrete sampling scheme, where sampled points are relatively independent, the key idea of DSConv is that later sampling positions depend on the offsets accumulated from earlier ones. If the vertical increment of the k-th sampled point is denoted by δy_k and the horizontal base step by Δx_k, then the snake-like sampling position can be written as

$$\tilde{p}_k = p + \left(\Delta x_k, \sum_{t=1}^{k} \delta y_t \right).$$

This formulation shows that DSConv no longer treats the sampled points as independent discrete positions, but lets them participate in feature extraction along a gradually unfolding trajectory. For retinal vessel segmentation, that matters because vessel direction, bifurcation, and local width variation are all closer to a continuous geometric form than to a scattered set of isolated points.

In the present implementation, this continuous trajectory is not confined to local single-scale sampling, but is further embedded into multi-scale aggregation. If the input feature at the l-th scale is denoted by x_l, the linearly projected representation by $P_l(x_l)$, and the corresponding weights by $A_{l,k}$ and $m_{l,k}$, then multi-scale dynamic snake aggregation can be summarized as

$$\mathrm{MSDSConv}(p) = \sum_{l=1}^{L}\sum_{k=1}^{K} A_{l,k} m_{l,k}\, P_l\big(x_l(\tilde{p}_{l,k})\big).$$

Compared with the multi-scale modulated deformable form used in the previous stage, the crucial change here is not merely that multi-scale aggregation continues, but that the formerly discrete cross-level sampled positions are replaced by a sequence of positions unfolding along a continuous trajectory. The model therefore no longer seeks only a number of isolated responses when accessing features across levels, but instead aggregates along a path that is more consistent with the geometry of the vessel itself. Figure 9 shows the combined structure of MS-DSConv and Coordinate Attention.

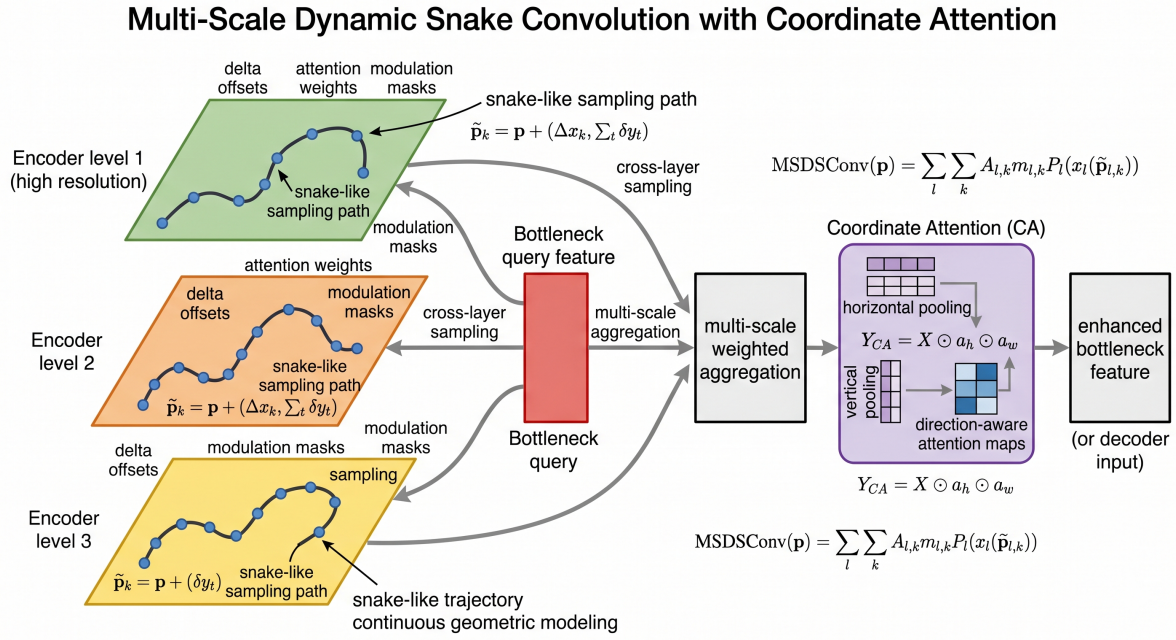


Figure 9: Structure of MS-DSConv combined with Coordinate Attention. The sampled points in the multi-scale branch unfold along a continuous snake-like trajectory; after cross-level weighted aggregation, Coordinate Attention further recalibrates the spatial response.

On that basis, the thesis uses MS-Modulated + CA as the direct comparison model and further constructs MS-DSConv + CA, in order to examine whether introducing continuous geometric trajectory modeling on top of multi-scale sampling and spatial attention can further improve the recovery of thin vessels and the integrity of the overall structure. The corresponding results are shown in Table 7.

As shown in Table 7, introducing Dynamic Snake Convolution on top of MS-Modulated + CA leads to clear gains in SE, AUC, and F1-score, with SE increasing to 0.8083, AUC increasing to 0.8949, and F1-score rising to 0.8020. This indicates that, once the sampling mechanism shifts from discrete point-

Table 7: Comparison between MS-Modulated + CA and MS-DSConv + CA

Model	SE	SP	ACC	AUC	PR	F1-score
MS-Modulated + CA	0.7697	0.9857	0.9673	0.8777	0.8324	0.7973
MS-DSConv + CA	0.8083	0.9815	0.9666	0.8949	0.7997	0.8020

based aggregation toward continuous trajectory modeling, the model gains a noticeably stronger ability to recover thin vessels, disconnected regions, and structurally continuous topology. At the same time, SP and PR decrease relative to the previous stage, and PR falls to 0.7997. In other words, stronger geometric continuity also makes the model more willing to label potential vessel pixels, which in turn increases some false positives and small noisy regions.

Taken together, the results of this section show that structural representation can be strengthened further on top of the multi-scale modeling line. The key additions are explicit positional attention and continuous geometric trajectory constraints. MS-Modulated + CA improves thin-vessel detection while preserving reasonable overall balance, whereas MS-DSConv + CA pushes structural continuity modeling further and achieves the best overall result at this stage. At the same time, its lower PR indicates that some false-positive regions still remain. This is why the next section turns to output refinement and loss-based constraints in order to improve the cleanliness and stability of the final result.

3.6 Post-processing and Loss-function Exploration

In the previous section, MS-DSConv + CA already achieved the best overall performance at the current stage, showing that the mainline built around multi-scale modeling, positional attention, and continuous geometric trajectory constraints can indeed strengthen the modeling of thin vessels and structural continuity. At the same time, however, a new problem became more visible: although this model performs strongly in SE, AUC, and F1-score, its PR still remains below 0.8. This means that, while the model becomes more aggressive in recovering thin vessels and continuous structures, it also retains some small noisy connected components and false-positive regions. The next step, therefore, is to deal more directly with these residual errors, first at the output level and then at the level of the loss function.

The first attempt in this direction is post-processing. Unlike the structural modifications in the previous sections, post-processing does not change the backbone network and does not directly affect feature extraction. Instead, it applies rule-based correction to the binarized prediction result. In retinal vessel segmentation, one common problem is the appearance of small, spatially isolated connected components, which often come from background noise, local reflections, or false positives. The thesis therefore adopts an area-threshold-based connected-component filtering strategy, with the aim of removing clearly unreasonable small regions while preserving the main vascular structure as much as possible. If a connected component in the prediction is denoted by C_i, its pixel area by $|C_i|$, and the area threshold by X, then the post-processing rule can be written as

$$\hat{Y}_{\text{post}}(C_i) = \begin{cases} 1, & |C_i| \geq X, \\ 0, & |C_i| < X. \end{cases}$$

Here, $X = 0$ means that no post-processing is applied; as X increases, more small connected components are removed. Figure 10 shows how the main metrics change under different thresholds, and Table 8 reports the corresponding results.

Table 8 shows that, without post-processing, namely at $X = 0$, the model already has the strongest vessel detection ability, with SE = 0.8083, AUC = 0.8949, and F1-score = 0.8020, but PR is only 0.7997. As the

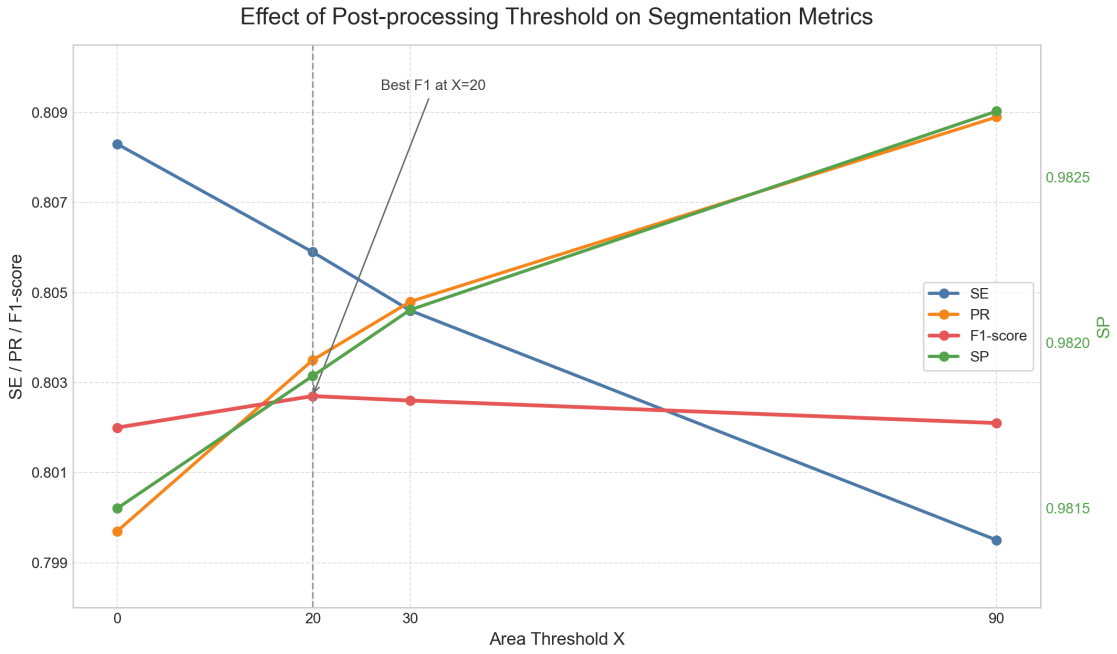


Figure 10: Influence of the post-processing area threshold on the main metrics. The left vertical axis shows SE, PR, and F1-score, while the right vertical axis shows SP, allowing the trends of different metrics to be viewed more clearly.

Table 8: Results of MS-DSConv + CA under different post-processing thresholds

Model	SE	SP	ACC	AUC	PR	F1-score
MS-DSConv + CA ($X = 0$)	0.8083	0.9815	0.9666	0.8949	0.7997	0.8020
MS-DSConv + CA ($X = 20$)	0.8059	0.9819	0.9668	0.8939	0.8035	0.8027
MS-DSConv + CA ($X = 30$)	0.8046	0.9821	0.9669	0.8933	0.8048	0.8026
MS-DSConv + CA ($X = 90$)	0.7995	0.9827	0.9670	0.8911	0.8089	0.8021

threshold rises from 0 to 20, 30, and 90, SP, PR, and ACC gradually increase, while SE and AUC gradually decrease. This indicates that post-processing does remove some small noisy regions, but it also inevitably removes some genuine thin vessels. A closer comparison shows that the highest F1-score, 0.8027, is obtained at $X = 20$, which suggests that this threshold provides a better balance between suppressing false positives and preserving fine vascular structures. As the threshold becomes larger still, the result becomes cleaner and more conservative, but the ability to retain thin vessels declines accordingly.

The thesis then asks whether structural continuity can be strengthened from the loss-function side. Unlike output correction, clDice is concerned with how connectivity information at the skeleton level can be incorporated into the optimization target during training [19]. Keeping the MS-DSConv + CA backbone unchanged, the thesis adds clDice on top of the original base loss in order to strengthen the modeling of skeleton consistency and connectivity in slender structures. If the prediction is denoted by P and the ground truth by G, clDice can be written as

$$\mathrm{clDice}(P, G) = \frac{2\, T_{\mathrm{prec}}(P, G)\, T_{\mathrm{sens}}(P, G)}{T_{\mathrm{prec}}(P, G) + T_{\mathrm{sens}}(P, G)},$$

where T_{prec} and T_{sens} denote topology-aware precision and topology-aware sensitivity defined on the basis of skeleton overlap. In the present implementation, clDice is applied only to the highest-resolution output branch and is optimized jointly with the original loss as an additional term. For direct comparison with the previous part, the thesis again evaluates the model under different area thresholds X, and the results are shown in Table 9.

Table 9: Results after adding clDice under different post-processing thresholds

Model	SE	SP	ACC	AUC	PR	F1-score
MS-DSConv + CA + clDice ($X = 0$)	0.8110 ± 0.0477	0.9806 ± 0.0052	0.9661 ± 0.0041	0.8958 ± 0.0220	0.7944 ± 0.0500	0.8003 ± 0.0243
MS-DSConv + CA + clDice ($X = 20$)	0.8079 ± 0.0484	0.9810 ± 0.0052	0.9663 ± 0.0041	0.8945 ± 0.0223	0.7976 ± 0.0500	0.8004 ± 0.0244
MS-DSConv + CA + clDice ($X = 30$)	0.8061 ± 0.0486	0.9812 ± 0.0052	0.9663 ± 0.0041	0.8937 ± 0.0224	0.7990 ± 0.0499	0.8002 ± 0.0245
MS-DSConv + CA + clDice ($X = 90$)	0.8002 ± 0.0498	0.9817 ± 0.0051	0.9663 ± 0.0042	0.8910 ± 0.0230	0.8025 ± 0.0496	0.7989 ± 0.0243

Table 9 shows that, after clDice is added, the model generally achieves slightly higher SE and AUC under different thresholds. For example, at $X = 0$, SE rises to 0.8110 and AUC to 0.8958; at $X = 20$, SE reaches 0.8079 and AUC 0.8945. This suggests that clDice, as an auxiliary loss aimed at topological consistency in slender structures, can indeed strengthen the model's sensitivity to thin vessels and continuous structures to some extent. However, compared with the results without clDice in the previous part, this modification does not lead to better overall performance. At the best post-processing threshold $X = 20$, the F1-score after adding clDice is 0.8004, lower than the 0.8027 obtained without it, and PR also drops from 0.8035 to 0.7976. In other words, although clDice moves in the right direction by strengthening structural continuity, it also introduces more false-positive regions and weakens the overall balance.

Taken together, this section shows that post-processing and clDice represent two different kinds of supplementary strategy: the former corrects local false positives at the output level, whereas the latter tries to strengthen structural continuity through the optimization objective itself. At the present stage of this study, the main limitation is not that topological constraints are entirely missing, but that local false positives produced by a high-sensitivity structural model still need to be corrected effectively. Under the current experimental conditions, post-processing provides a more direct and stable gain, whereas clDice remains a reasonable exploration but does not outperform the post-processing-based solution as a whole. The thesis therefore retains post-processing as an important part of later result refinement, while treating clDice as an exploratory but unretained attempt.

3.7 Chapter Summary

This chapter has followed the experimental progression from early exploration to later structural optimization within the unified nnU-Net framework. The early experiments show that, on top of nnU-Net's existing preprocessing and training pipeline, extra input enhancement and loss-function adjustment can influence the training process and some local metrics, but they do not produce sufficiently stable gain in the final result. In particular, under this early-stage setting, additional input-side preprocessing does not deliver the expected improvement. From that point onward, the study moves away from local training-side adjustment and toward structural improvement of the network itself.

The later stages of the chapter begin with PA-Net module migration as a first round of structural exploration. Through full migration, single-module ablation, and the control experiment on AFFM and deep supervision, the thesis carries out an initial screening of how different modules adapt to the present framework. The overall result is that the complete PA-Net migration does not yield a better solution, AFFM does not provide stable gain, and LPT shows a clearer value for continued optimization. From there, the structural mainline gradually converges on LPT and continues through deformable sampling, modulation, multi-scale aggregation, positional attention, and continuous geometric trajectory modeling.

As structural continuity modeling becomes stronger, the model also begins to expose a corresponding problem in the form of false positives, even when SE, AUC, and F1-score improve. This is why the second half of the chapter turns to supplementary exploration from the perspectives of output refinement and loss-based constraints. The results show that area-threshold post-processing can correct local false positives more directly and provide a more stable overall gain, whereas clDice, although reasonable in direction, does not produce a better overall result under the current experimental conditions. What is finally retained in this chapter is therefore not one isolated module, but a mainline solution formed through gradual screening together with its necessary output-side refinement strategy.

In that sense, this chapter is really about how the model was modified step by step. If Chapter 2 explains what these methods are, Chapter 3 explains how they are selected, combined, and implemented in the present work. The next chapter will examine the models formed here from the perspectives of overall results, metric trade-offs, and structural evolution.

4 Experimental Results and Analysis

4.1 Overall Comparison of Results

This section compares the representative models selected from different stages of the study in order to establish a common point of reference. It includes both the early contrast from direct PA-Net migration and the key models that emerged later along the LPT mainline, so that the overall trajectory of the performance changes can be seen more clearly. As noted earlier, the current AUC values are approximated from binarized predictions, so the comparison here considers SE, PR, SP, and F1-score together rather than relying on a single metric alone.

Table 10 makes it clear that the full PA-Net migration does not outperform the nnU-Net v2 baseline. Although PA-Net (Full) reaches a PR of 0.8570, which suggests a more conservative output with fewer false positives, its SE is only 0.7211, clearly lower than the baseline value of 0.7627, and its final F1-score also drops to 0.7782. In other words, the fully migrated model tends to preserve high-confidence vessel regions at the cost of losing many thin vessels and weak-response areas. The design ideas themselves are therefore not without value, but a direct transfer into the nnU-Net framework does not automatically produce a better

Table 10: Overall comparison of representative models in Chapter 4

Model	SE	SP	ACC	AUC	PR	F1-score
nnU-Net v2 Baseline	0.7627	0.9852	0.9659	0.8739	0.8238	0.7891
PA-Net (Full)	0.7211	0.9888	0.9659	0.8549	0.8570	0.7782
Deformable LPT (No AFFM, 4 point)	0.7634	0.9863	0.9672	0.8749	0.8371	0.7958
MS-Modulated (4 point)	0.7582	0.9870	0.9673	0.8726	0.8419	0.7954
MS-Modulated + CA	0.7697	0.9857	0.9673	0.8777	0.8324	0.7973
MS-DSConv + CA	0.8083	0.9815	0.9666	0.8949	0.7997	0.8020
MS-DSConv + CA ($X = 20$)	0.8059	0.9819	0.9668	0.8939	0.8035	0.8027

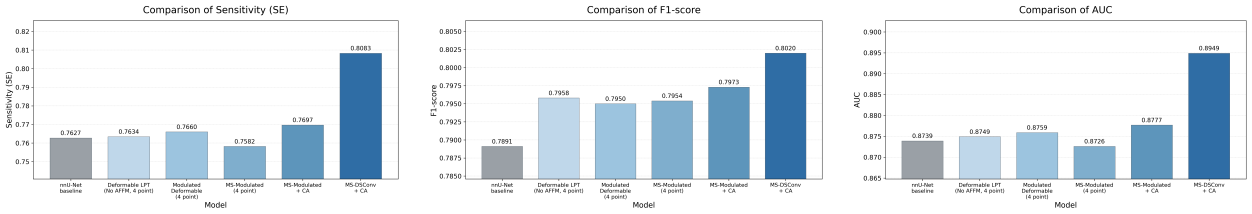
solution.

The later improvements along the LPT line, by contrast, show a more stable upward trend. Deformable LPT (No AFFM, 4 point) raises F1-score to 0.7958, indicating that more flexible sampling at the bottleneck does improve high-level vessel-structure modeling. MS-Modulated does not further increase SE, but it performs better in SP, ACC, and PR, which reflects a more balanced overall behavior. Adding CA on top of that raises SE, AUC, and F1-score again, suggesting that explicit directional and positional information still contributes positively to the recovery of slender vessels. A further step to MS-DSConv increases SE to 0.8083 and F1-score to 0.8020, showing that continuous geometric trajectory modeling is better matched to this task at the structural level.

The final result appears in the MS-DSConv + CA model with post-processing at $X = 20$, where F1-score reaches 0.8027, representing an improvement of 0.0136 over the baseline value of 0.7891. This gain is not produced by post-processing alone. It depends first on the stronger vessel-detection ability established through the earlier structural modifications and then on the correction of the low-PR problem through lightweight output filtering. Put differently, the final retained solution is not a case of one isolated module suddenly succeeding, but the outcome of multiple rounds of structural optimization and result screening acting together.

4.2 Analysis of Metric Trade-offs

From the perspective of retinal vessel segmentation, SE mainly reflects the detection of thin vessels and weak-response regions, PR more directly reflects the suppression of false positives, and F1-score represents the balance between the two. Model comparison therefore cannot rely on whether one metric is simply higher than another. What matters more is whether a structural modification pushes the model toward a more sensitive or a more conservative prediction pattern. The figures in this section focus on the later mainline models, namely the structures that evolved after the baseline along the LPT route and were eventually retained.



(a) SE comparison (b) F1-score comparison (c) AUC comparison

Figure 11: Changes in SE, F1-score, and AUC across representative models on the later mainline. The figure shows the key structural versions that emerged along the LPT route after the baseline.

Figure 11 first shows the overall changes in SE, F1-score, and AUC along the later mainline. Deformable

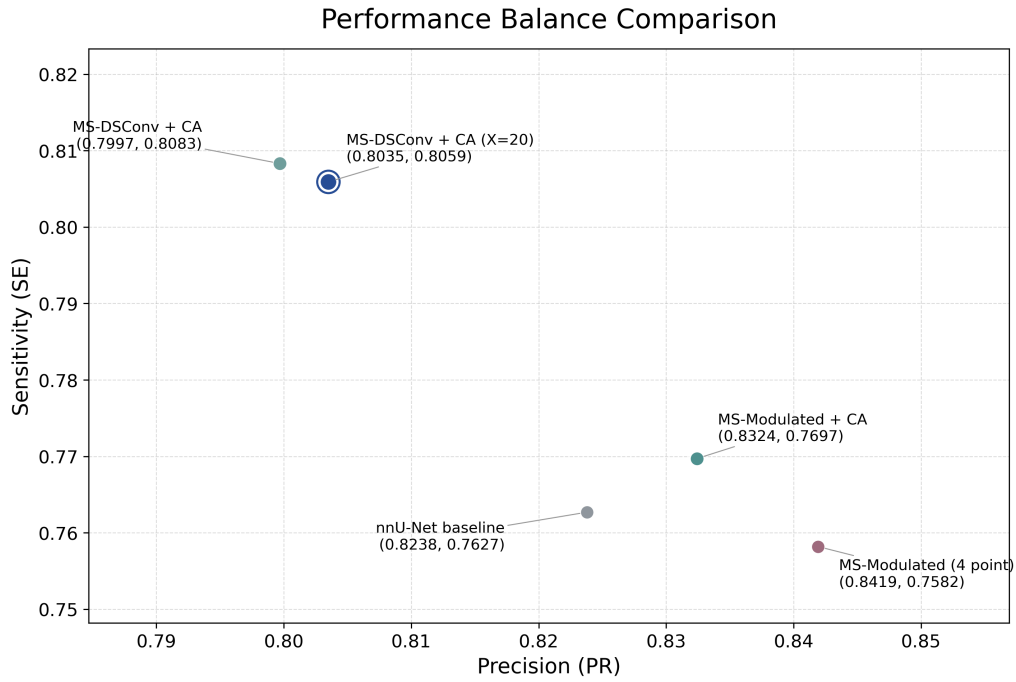


Figure 12: Performance balance between sensitivity and precision for the representative later-mainline models, including the final post-processed version.

LPT and Modulated Deformable are both more inclined to improve the detection of thin vessels. Deformable LPT raises F1-score to 0.7958 relative to the baseline and also slightly improves SE, which suggests that more flexible sampling already helps high-level structural modeling. Modulated Deformable pushes SE further to 0.7660 and also reaches a comparatively high AUC at this stage, indicating that the addition of a modulation mask makes the model more attentive to potential vessel responses. From the perspective of F1-score, however, this step does not bring a more obvious overall advantage than Deformable LPT, so it is better understood as a further move in the direction of stronger detection rather than a major improvement in balance.

The later half of the mainline is reflected more clearly in the SE-PR relationship shown in Fig. 12. MS-Modulated lies further to the right, indicating relatively high PR but comparatively limited SE. This suggests that cross-level multi-scale sampling is more helpful for suppressing background noise and maintaining a conservative prediction style. After CA is introduced, the point moves upward, which means that SE recovers while PR remains relatively high. Combined with Fig. 11 and Table 10, this step improves not only the balance relation itself, but also AUC and F1-score. In that sense, CA does not simply push false positives down further. More importantly, it strengthens the representation of vessel direction and position on top of the existing multi-scale feature interaction.

On this basis, MS-DSConv + CA shows an even clearer tendency to trade stronger structural continuity for stronger detection. Figure 11 shows that it reaches higher SE, F1-score, and AUC, with SE increasing to 0.8083, F1-score to 0.8020, and AUC to 0.8949. Figure 12, however, shows that these gains come with a drop in PR. In other words, DSConv makes the model more willing to identify potential vessel pixels. This is beneficial for recovering thin vessels and continuous structures, but it also inevitably introduces more small noisy regions.

The final point in Fig. 12 further clarifies the role of post-processing. For MS-DSConv + CA, post-processing does not fundamentally alter the model's high-sensitivity character. Instead, it removes some isolated noise, raising PR from 0.7997 to 0.8035 while causing only a small loss in SE. Together with Table

10, this also explains why the post-processed version reaches the highest F1-score of the current stage, namely 0.8027. The strength of the final retained solution, then, does not lie in making one single metric absolutely optimal, but in reaching a more suitable compromise between vessel detection, false-positive control, and overall balance.

4.3 Analysis of the Structural Evolution Path

If the previous two sections focus on what the models achieved, this one asks why the final route settled where it did. Chapter 3 shows that the work did not move toward a fixed endpoint by steadily adding modules. The direction only tightened after several rounds of trial and screening. The early PA-Net migration made one point fairly clear: directly transplanting an existing design did not produce a better model under the present framework. The part that kept justifying further work was the bottleneck LPT. Once that became clear, the attention moved away from decoder-side fusion and back to the vessel structure itself, especially how high-level features could better describe curved, thin, and heavily branched vessels.

Seen from that angle, Deformable LPT and Modulated Deformable first deal with a fairly direct problem: the sampling mechanism is not flexible enough. Fixed local aggregation is not weak in ordinary regions, but it becomes less convincing when vessel trajectories change quickly and scale differences become obvious. Deformable sampling loosens the reliance on fixed positions. The added modulation mask then pushes the model to respond more readily to potential vessel regions. In the results, the earliest sign of this shift is the rise in SE. Even so, this stage still feels like a step toward more active vessel detection rather than a full solution, because the recovery of shallow details is not yet stable enough.

The emphasis changes again from Modulated Deformable to MS-Modulated and then to MS-Modulated + CA. At that point, the issue is no longer just whether sampling is flexible, but whether information from different levels can work together more naturally. Multi-scale sampling matters not simply because more feature maps are involved, but because shallow details are brought back into high-level structural judgment. That reduces the model's dependence on deep semantic abstraction alone when it has to infer vessel trajectories. CA builds on that adjustment rather than changing the route itself. Its role is smaller and more specific: it makes the model more responsive to directional and positional cues. By then, the goal is no longer only to push detection upward. It is to make structural expression steadier without letting the prediction become too messy.

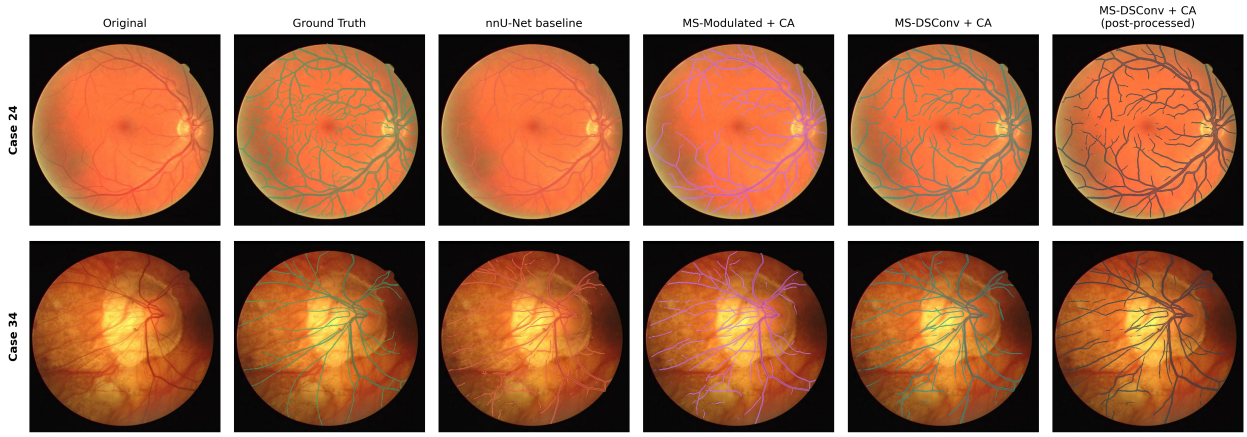


Figure 13: Visual comparison of segmentation results for representative cases. From left to right: original image, ground truth, nnU-Net baseline, MS-Modulated + CA, MS-DSConv + CA, and its post-processed result.

The two cases in Fig. 13 are better read as examples of different prediction styles than as proof of a strict ranking among the intermediate models. Taken broadly, the baseline remains more conservative, while the later mainline models respond more actively to potential vessel regions. In Case 24, MS-Modulated + CA does not show a clearly stronger recovery of fine branches than the baseline. On the whole, it looks more restrained. Once the model moves to MS-DSConv + CA, it keeps more vessel regions and reaches higher overlap with the annotation. Case 34 points to something slightly different. The intermediate models do not improve in a neat linear sequence. Both MS-Modulated + CA and MS-DSConv + CA are better than the baseline, but not in exactly the same way. The former looks somewhat cleaner, while the latter preserves some vessel regions more aggressively and ends up with slightly higher overlap. The last column shows post-processing in a fairly plain role: it removes scattered small regions without changing the main structural tendency already visible in the previous result.

That is why DSConv was retained. The reason is not just that a few metrics rise again. It also fits the logic of the steps before it. Multi-scale sampling and CA provide a more suitable feature basis, then DSConv carries the preference for thin and continuous structures down to the local modeling level. Under those conditions, the model is more willing to produce continuous responses when vessels are strongly curved or densely branched. The trade-off is obvious enough. More scattered noise appears, which is why post-processing still matters. In the end, the convergence to MS-DSConv + CA together with post-processing is not really about keeping the version with the largest number of modules. It is about keeping the combination that best matches the problem the thesis is actually trying to solve.

4.4 Supplementary Discussion on Post-processing and clDice

The previous section already explains why the main line ends at MS-DSConv + CA. That does not mean every remaining problem disappears at that point. It simply means that this is the stage at which the gains in structural representation and thin-vessel recovery become most worth keeping. At the same time, the remaining weakness is easier to see. Figure 12 and Table 10 show that MS-DSConv + CA already reaches relatively high SE, AUC, and F1-score, while PR still lags behind. By this stage, the limiting factor is less the recovery of the major vessel trunks, and less a simple lack of topology-aware constraints, than the scattered noise and local false positives that appear when vessel regions are recovered more aggressively.

That is why post-processing stays in the final solution. It does not replace the structural changes that came before it. Instead, it makes a light correction at the output stage, where the current error pattern is easiest to handle. For a high-sensitivity model such as MS-DSConv + CA, the main issue is usually not that the vascular tree is grossly wrong. The more common problem is the presence of small connected components near the boundary that remain isolated from the main structure. Errors like these are easier to remove at the output level than to suppress by pushing the backbone to become more conservative again. Table 8 reflects this quite clearly. At $X = 0$, the model reaches an F1-score of 0.8020 with PR = 0.7997. When an area threshold of $X = 20$ is applied, PR rises to 0.8035 and F1-score to 0.8027, while SE falls only slightly from 0.8083 to 0.8059. The improvement is modest, but it addresses exactly the kind of error that still matters here. That is enough to justify keeping post-processing in the final version.

clDice is harder to justify in the same way. It was not dropped because it did nothing. The issue is that the kind of gain it brings does not line up well with the main problem at this stage. As shown in Table 9, adding clDice does lead to slightly higher SE and AUC in some settings. At $X = 0$, for example, SE rises from 0.8083 to 0.8110 and AUC from 0.8949 to 0.8958. This suggests that introducing skeleton-level connectivity into the training objective really does make the model more responsive to slender structures. But the change does not improve the overall balance. Under the more representative setting of $X = 20$,

F1-score falls to 0.8004 from 0.8027, and PR drops from 0.8035 to 0.7976. In practice, clDice seems to make the model more sensitive to potential vessel regions without helping enough on local false positives.

Seen together, post-processing and clDice reflect two different ways of dealing with what remains. Post-processing works at the output stage and cleans up local errors after they appear. clDice works during training and tries to promote structural continuity through the objective itself. Along the present main line, once the model reaches MS-DSConv + CA, structural continuity is no longer the weakest part of the system. The more pressing issue is how to tidy the predictions produced by a high-sensitivity model. In that setting, post-processing gives the more direct and more reliable gain. clDice is still a useful exploration, but it does not outperform the post-processing-based version as a whole. The final retained combination is therefore MS-DSConv + CA with light post-processing, while clDice remains an informative attempt rather than part of the final model.

4.5 Chapter Summary

This chapter brought the stagewise models back under a common reference and looked at more than raw ranking. The question throughout was also how those results came into being. Under the present framework, a full PA-Net migration does not move past the baseline. The path that keeps yielding useful gains is the one built around LPT. As deformable sampling, modulation, multi-scale modeling, positional attention, and continuous geometric trajectory modeling are added one after another, the model becomes better at handling thin vessels and structural continuity. That is why MS-DSConv + CA ends up as the later-stage structure worth keeping.

Still, Chapter 4 is not really about naming a single winning module. Once the metrics and the qualitative results are read together, the pattern is fairly consistent. The later structural changes keep negotiating the same balance: stronger detection, fewer false positives, and steadier overall behavior. Deformable LPT and Modulated Deformable lean more toward finding as many vessel regions as possible. Multi-scale modeling and CA make the output more stable. By the time the model reaches MS-DSConv + CA, thin and continuous structures are recovered more effectively, but local false positives become harder to ignore. So what remains at the end is not one isolated best component. It is a mainline solution that gradually emerges through repeated comparison.

The discussion of post-processing and clDice sharpens that point. At this stage, the real issue is no longer whether vessels can be segmented at all, but how to clean up the extra small noisy regions left behind by a high-sensitivity prediction. Area-threshold post-processing addresses that problem directly, so it brings the more immediate and more stable gain. clDice makes sense in principle and does make the model more responsive to slender structures, but it does not improve the overall balance. In the end, the solution retained in Chapter 4 is a result-refinement scheme built around MS-DSConv + CA with light post-processing.

5 Conclusions and Future Work

5.1 Overall Conclusions

This thesis investigates retinal vessel segmentation within the nnU-Net framework. The main concern throughout is how to improve the recovery of thin vessels, make structural continuity more stable, and reduce local false positives under a unified experimental setting. Retinal vessel segmentation may look like a pixel-level classification problem, but the difficulty does not stop at separating foreground from background. A large part of the challenge is preserving thin, curved, and densely branched vessels under complicated imag-

ing conditions. For that reason, the thesis does not step away from an existing strong baseline to build a completely new framework. It keeps nnU-Net as the common starting point and asks, under relatively stable conditions, what different structural changes and result-refinement strategies actually add.

The experiments do not unfold around one single baseline all the way through. Instead, the work moves in two stages: an early exploration stage and a later main-experiment stage. Both stages start from nnU-Net, but the early one is more concerned with training-side exploration, especially non-structural changes such as input enhancement and loss adjustment. That stage is mainly used to judge what these measures can still add on top of nnU-Net's existing pipeline. The more specific conclusion is that the extra input-side preprocessing does not bring clear gain. A more plausible explanation is that nnU-Net already provides a mature internal preprocessing pipeline, so additional manual processing before the image enters the network does not produce the expected benefit. On that basis, the focus later shifts back to the network structure itself.

In the later main-experiment stage, the thesis establishes a new nnU-Net baseline under a unified data-organization scheme and training protocol, and uses it as the shared reference for the structural comparisons that follow. The first round of screening begins with PA-Net module migration, together with full migration, module ablation, and related control experiments to see which modifications actually fit the current framework. After that, the main line narrows gradually toward LPT and continues through deformable sampling, modulation, multi-scale modeling, positional attention, and continuous geometric trajectory modeling. The structure that eventually emerges from this process is the later-stage mainline version centered on MS-DSConv + CA.

The thesis then turns to post-processing and clDice as supplementary forms of result refinement. As structural continuity modeling becomes stronger, the model gains sensitivity and improves its recovery of thin vessels, but it also starts to leave behind more local false positives. Area-threshold post-processing addresses that issue more directly because it removes scattered false-positive regions while improving the overall balance without clearly changing the main structural trend. clDice makes sense in principle and does increase the model's sensitivity to slender structures, but under the present experimental conditions it does not produce better overall performance. Seen across the whole thesis, the final retained outcome is therefore not one isolated module, but a complete solution that gradually takes shape through repeated comparison, screening, and supplementary refinement: a retinal vessel segmentation method centered on MS-DSConv + CA and supported by light post-processing.

5.2 Main Findings

The main findings of this thesis fall into four closely connected parts. They are not meant to stand as isolated observations. Read together, they show how the study moved from preliminary exploration to a structure-centered solution built on nnU-Net, and why the final retained scheme converged on MS-DSConv + CA with light post-processing.

One concerns the overall experimental framework. The experiments are divided into an early exploration stage and a later main-experiment stage, both within the same nnU-Net framework. That separation makes it easier to see what each stage is actually answering. The early experiments are mainly used to test non-structural changes such as input enhancement and loss adjustment, whereas the later stage rebuilds a baseline under a unified data-organization scheme and training protocol and then uses it as the reference for structural comparison. Taken together, these results support a fairly specific conclusion: under the early exploration setting, extra input-side preprocessing does not lead to clear gain. The more convincing explanation is not that such adjustment is meaningless in principle, but that nnU-Net already includes a relatively complete

internal preprocessing pipeline. Under those conditions, extra manual processing before the image enters the network does not offer the improvement one might expect. This leaves structural modification as the more worthwhile direction for later work.

Another comes from the first round of structural screening based on PA-Net module migration. Full migration, module ablation, and related control settings were compared systematically. The outcome is fairly clear. A full PA-Net migration does not produce a better solution in the present framework, and the gain from AFFM is not stable. The component that continues to justify further optimization is the LPT placed at the bottleneck. Under the present task setting and experimental conditions, stronger high-level structural modeling turns out to be a more promising direction than directly adopting a complete external design.

A third finding follows from the later mainline itself. Once LPT is taken as the mainline direction, the work continues through deformable sampling, modulation, multi-scale modeling, positional attention, and continuous geometric trajectory modeling, and gradually forms the core structure used in the later main experiments. Both the overall results and the metric analysis suggest that this line steadily improves the model's ability to represent thin vessels and continuous structures. In the later main experiments, MS-DSConv + CA gives the strongest overall performance. After a light post-processing step with an area threshold of $X = 20$, the F1-score reaches 0.8027, which is 0.0136 higher than the later-stage nnU-Net baseline result of 0.7891. This supports the view that a modeling strategy built around the continuous geometric trajectory of slender structures is effective for the retinal vessel segmentation task studied here.

The last finding comes from the comparison between post-processing and clDice as supplementary refinement strategies. Area-threshold post-processing removes scattered false-positive regions more directly and improves the overall balance while largely preserving the main structural trend. clDice makes sense in direction and does increase the model's sensitivity to slender structures, but under the present experimental conditions it does not produce better overall performance. What is finally retained, then, is not a single standout module, but a complete scheme that takes shape through repeated screening: a retinal vessel segmentation method centered on MS-DSConv + CA and supported by light post-processing.

5.3 Limitations and Possible Improvements

Although the thesis develops a fairly clear later-stage mainline solution within the nnU-Net framework, several limitations remain.

One of them lies in how the study handles nnU-Net's internal preprocessing pipeline. The early experiments show that extra input-side preprocessing before the image enters the network does not bring obvious gain, which suggests that nnU-Net already has a fairly complete preprocessing mechanism of its own. Even so, the thesis does not go further and unpack the role of each internal preprocessing step for the present task, nor does it attempt more targeted adjustment of normalization, channel organization, or low-contrast enhancement for retinal vessel segmentation. At this stage, the explanation for why extra preprocessing fails to help remains tied mostly to observed behavior rather than a finer-grained analysis of the pipeline itself.

Another limitation is the scope of validation. The early exploration stage and the later main-experiment stage both establish relatively clear comparison references, and the later experiments also compare structures under a unified protocol. Even so, the findings are still built mainly on the public data conditions used here. That helps keep the horizontal comparison interpretable, but it also limits how much can be said about generalization. Whether the retained solution would keep the same advantage across data from different imaging devices, different pathological conditions, or different annotation styles still needs broader verification.

There is also a methodological limitation in the final retained solution. It still relies on light post-processing to correct local false-positive regions. This means that, although the structural mainline improves

thin-vessel recovery and continuous-structure representation, false-positive control has not been handled entirely in an end-to-end way. In the current result, the final balance comes partly from the structural design itself and partly from additional filtering at the output stage. That combination works, but it still leaves open the question of whether this correction ability could be absorbed more directly into the network structure or the training objective.

The last limitation concerns interpretation. The present conclusions rely mainly on metric changes, qualitative comparison, and contrast among different structures. That is enough to show which modifications help and which ones are not retained, but it does not fully explain why some of these patterns appear. For example, it is still unclear why certain structures produce more sensitive or more conservative predictions on particular cases, or how DSConv interacts internally with multi-scale sampling and positional attention. These questions would need more detailed intermediate-feature analysis and stronger interpretability support. The qualitative results point in the same direction. Even though the retained models recover the main vessel structures more effectively and improve thin-vessel detection to some extent, they still miss some fine branches, leave local discontinuities, and remain weak in some low-response regions when compared with the ground truth. So there is still clear room for improvement in fine-grained structural completeness. For that reason, the conclusions of this thesis are better read as stage-specific findings under the current experimental conditions than as a complete account of every mechanism involved.

5.4 Future Work

One obvious next step is to look more closely at nnU-Net's internal preprocessing pipeline from the perspective of this task. The early experiments show that extra input-side preprocessing does not bring obvious gain, but that does not mean the preprocessing side has nothing left to offer. A more useful follow-up would be to examine the role of the internal preprocessing steps one by one in retinal vessel segmentation, and then explore refinements that better match the task, especially in normalization, channel organization, and the handling of low-contrast regions.

It would also be worth pushing false-positive control further toward an end-to-end solution. In the retained scheme, local false positives are corrected effectively by light post-processing, but that correction still happens after the network has already produced its output. If a similar cleaning effect could be brought more directly into the network structure or the training objective, the model might keep its ability to recover thin vessels while depending less on extra rule-based filtering. That would make the overall method more self-contained.

The architecture itself also leaves an interesting question open. The retained mainline already performs well in thin-vessel recovery and continuous-structure representation, but the multi-scale sampling and related structural enhancement remain concentrated mainly on the bottleneck and encoder-side feature aggregation, without a corresponding mirrored design on the decoder side. So while the model still keeps the basic encoder-decoder framework of U-Net, the actual distribution of the main modifications has become somewhat asymmetric. A natural extension would be to test whether a more coordinated symmetric design can be built without giving up the strengths of the current mainline, and then examine how such a symmetric variant affects performance, structural representation, and result stability. This matters not just for architectural neatness. It may also help clarify which parts of the present mainline are truly doing the heavy lifting.

The retained solution should also be tested under broader data conditions. At present, the main conclusions still rest on the public datasets used here and on a unified experimental protocol. Extending the evaluation to data from different imaging devices, different pathological conditions, and different annotation styles would give a clearer picture of the model's generalization ability and its practical value.

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